

Unit 2

ATOMS, MOLECULES AND STOICHIOMETRY

77 items · 263 marks

Syllabus sections in this unit

- 2.1 Relative masses of atoms and molecules
- 2.2 The mole and the Avogadro constant
- 2.3 Formulas
- 2.4 Reacting masses and volumes (of solutions and gases)

Unit 2: Atoms, molecules and stoichiometry

1. M/J 26/P21/Q1/e

(e) Compound **X** contains sulfur and chlorine only.

100.0 g of **X** contains 47.49 g of sulfur. The relative molecular mass, M_r , of **X** is 135.2.

Determine the molecular formula of **X**. Show your working.

molecular formula of **X** = [3]

2. M/J 26/P22/Q4/b(ii,iii)

(b) The structure of compound **X** is shown in Figure 4.2.

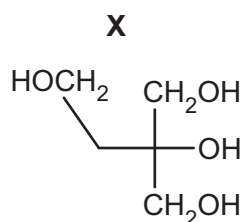


Figure 4.2

The molecular formula of **X** is $C_5H_{12}O_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of 0.500 mol dm^{-3} acidified $K_2Cr_2O_7(aq)$.

(ii) Calculate the amount, in mol, of $K_2Cr_2O_7(aq)$ used in the reaction.

amount = mol [1]

(iii) Calculate the amount, in mol, of **X** present in the 0.255 g sample.

amount = mol [1]

3. M/J 26/P24/Q1/b

1 Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

(b) Deduce the formula of aluminium nitrate(V).

[1]

4. O/N 25/P21/Q3/a

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

[1]

5. O/N 25/P23/Q3/a

- 3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

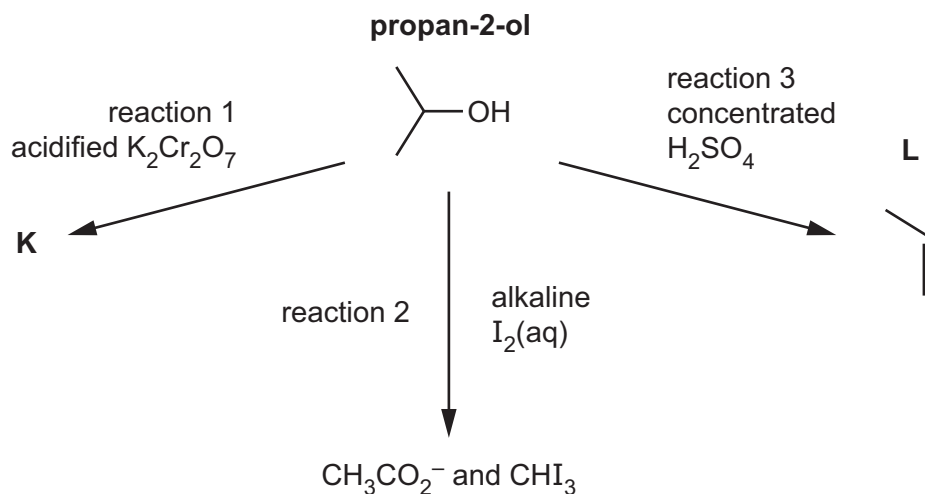
(a) Define empirical formula.

[1]

6. O/N 25/P24/Q3/b(iv)

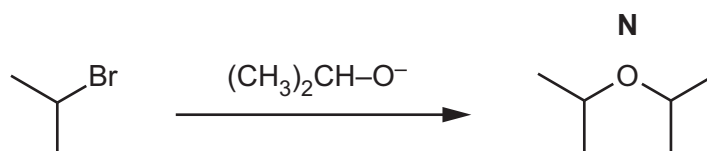
- 3 Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

**Fig. 3.1**

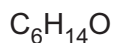
- (b) Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH}-\text{O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

**Fig. 3.3**

- (iv) **N** is also added to petrol to make it burn more smoothly.

Construct an equation for the complete combustion of **N**, $\text{C}_6\text{H}_{14}\text{O}$.



[1]

7. M/J 25/P21/Q2/b(i)

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

(b) Fig. 2.1 shows the mass spectrum of the sample of iron.

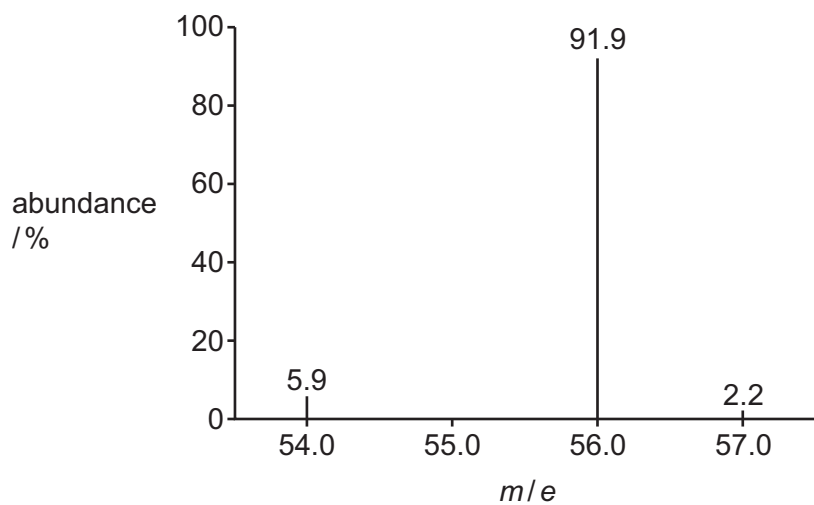


Fig. 2.1

- (i)** Use Fig. 2.1 to calculate the relative atomic mass, A_r , of iron to **one** decimal place.

Show your working.

A_r of iron = [2]

8. M/J 25/P21/Q4/a,b(i,ii)

4 The structure of vitamin C is shown in Fig. 4.1.

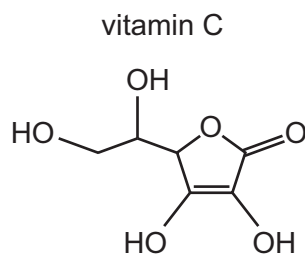


Fig. 4.1

- (a)** Deduce the empirical formula of vitamin C.

[1]

- (b)** The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

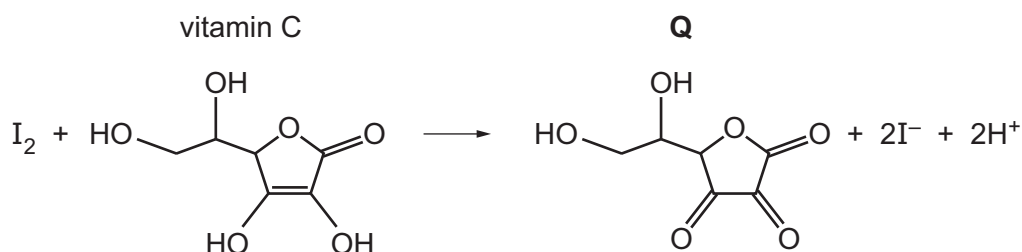


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm^3 of vitamin C solution. 5.00 cm^3 of this vitamin C solution is added to a flask with approximately 150 cm^3 of water and an indicator.

Exactly 28.40 cm^3 of $5.00 \times 10^{-4}\text{ mol dm}^{-3}\text{ I}_2(\text{aq})$ reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

- (i) Calculate the amount, in mol, of $\text{I}_2(\text{aq})$ added to the flask in this titration.

amount of $\text{I}_2(\text{aq}) =$ mol [1]

- (ii) Use your answer to (b)(i) to calculate the mass, in g, of vitamin C in the tablet. Show your working.

(If you were unable to calculate a value for the amount of $\text{I}_2(\text{aq})$ in **(b)(i)**, use the value $2.64 \times 10^{-4} \text{ mol}$. This is **not** the correct value.)

mass of vitamin C in tablet = _____ g [2]

9. M/J 25/P22/Q2/a

2 A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(a) (i) Define relative atomic mass.

[2]

(ii) Table 2.1 shows the abundances of two of the isotopes present in the sample of iron.

Table 2.1

isotope	relative isotopic mass	abundance / %
^{54}Fe	53.9	6.0
^{56}Fe	55.9	91.9

Use Table 2.1 to calculate the relative isotopic mass of the third isotope of iron in the sample. Show your working.

relative isotopic mass =

[2]

10. M/J 25/P22/Q5/a,b(i,ii)

5 The structure of vitamin C is shown in Fig. 5.1.

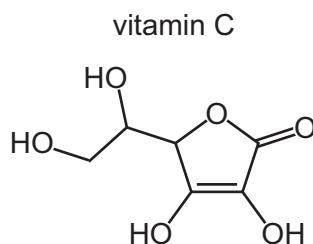


Fig. 5.1

- (a)** Deduce the empirical formula of vitamin C.

[1]

- (b)** The mass of vitamin C present in 150.0g of lemon is found in an experiment.

stage 1 All the vitamin C in 150.0g of lemon is extracted and dissolved in water to make 100.0 cm³ of solution L.

stage 2 A 25.0 cm^3 sample of solution **L** is titrated with $5.00 \times 10^{-3}\text{ mol dm}^{-3}\text{ I}_2(\text{aq})$.

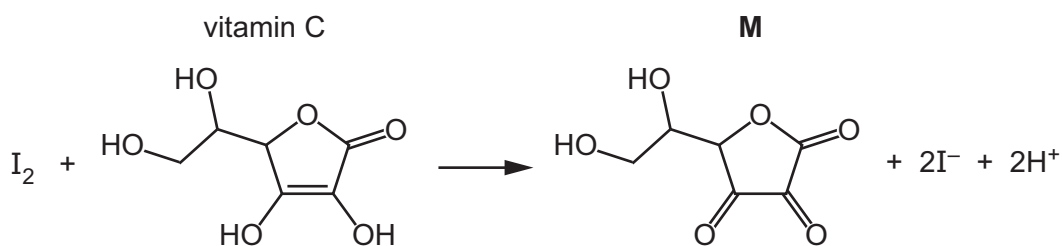


Fig. 5.2

Exactly 36.65 cm^3 of $\text{I}_2(\text{aq})$ reacts with the 25.0 cm^3 sample of solution L.

[M_r : vitamin C, 176]

- (i) Calculate the amount, in mol, of $\text{I}_2(\text{aq})$ added in the titration.

amount of $\text{I}_2(\text{aq}) =$ mol [1]

- (ii) Use your answer to (b)(i) to calculate the percentage by mass of vitamin C present in 150.0 g of lemon.

(If you were unable to calculate a value for the amount of $\text{I}_2(\text{aq})$ in **(b)(i)**, use the value $7.65 \times 10^{-4} \text{ mol}$. This is **not** the correct value.)

percentage by mass = % [2]

11. M/J 25/P23/Q5/a(i)

5 (a) **W** is a colourless liquid.

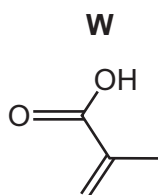


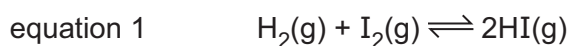
Fig. 5.1

(i) Deduce the empirical formula of **W**.

[1]

12. M/J 25/P24/Q3/a

3 $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



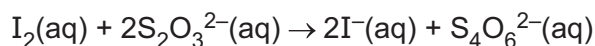
In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

(a) In experiment 1, $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ are combined.

Water is then added to the equilibrium mixture to produce 1.00 dm^3 of solution **A**.

The amount of $\text{I}_2(\text{aq})$ present in 1.00 dm^3 of solution **A** is found by titration with $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$.

Exactly 32.90 cm^3 of $0.200 \text{ mol dm}^{-3}$ $\text{Na}_2\text{S}_2\text{O}_3$ reacts with all of the I_2 in a 25.0 cm^3 sample of solution **A**.



(i) Calculate the amount, in mol, of $\text{S}_2\text{O}_3^{2-}$ that reacts in the titration.

amount of $\text{S}_2\text{O}_3^{2-} =$ mol [1]

(ii) Use your answer to (a)(i) to calculate the amount, in mol, of I_2 present in 1.00 dm^3 of solution **A**.

(If you were unable to obtain an answer in (a)(i), then use $9.42 \times 10^{-3} \text{ mol}$. This is **not** the correct answer.)

amount of I_2 in 1.00 dm^3 of solution **A** = mol [2]

13. M/J 25/P24/Q6/a(i,ii)

6 **P**, **Q** and **R** are three different hydrocarbon molecules.

(a) (i) Define empirical formula.

[1]

(ii) Hydrocarbon **P** contains 85.7% by mass of carbon.

Calculate the empirical formula of **P**.

Show your working.

empirical formula of **P** =

[2]

14. O/N 24/P21/Q1/c,d(i)

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(c) Table 1.2 shows the relative abundances of isotopes in a sample of an alloy containing rhodium and iridium only.

Table 1.2

isotope	relative isotopic mass	relative abundance in alloy
$^{103}_{45}\text{Rh}$	102.91	50.00
$^{191}_{77}\text{Ir}$	190.96	15.18
$^{193}_{77}\text{Ir}$	192.96	34.82

(i) Define relative isotopic mass.

[2]

(ii) Use Table 1.2 to calculate the relative atomic mass, A_r , of iridium in the alloy.

Give your answer to **two** decimal places.

relative atomic mass of iridium =

[2]

(d) Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$.

Show your working.

$x =$

[2]

15. O/N 24/P22/Q1/b(ii,iii)

1 Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

(b) Pelopium was the suggested name for a new element discovered in a mineral.

Pelopium was later found to be a mixture of niobium, Nb, and tantalum, Ta.

Only one naturally occurring isotope exists for each of Nb and Ta.

Table 1.1

isotope	relative isotopic mass	number of protons	number of neutrons
${}^{93}_{41}\text{Nb}$	92.91		
${}^{181}_{73}\text{Ta}$	180.95		

(ii) Define relative isotopic mass.

[2]

(iii) A sample of pelopium contains 90.9% by mass ${}^{93}_{41}\text{Nb}$ and 9.1% by mass ${}^{181}_{73}\text{Ta}$.

Calculate the theoretical relative atomic mass of pelopium based on these data and Table 1.1.

Give your answer to two decimal places.

Show your working.

theoretical relative atomic mass of pelopium =

[2]

16. O/N 24/P23/Q1/c,d(i)

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(c) Table 1.2 shows the relative abundances of isotopes in a sample of an alloy containing rhodium and iridium only.

Table 1.2

isotope	relative isotopic mass	relative abundance in alloy
$^{103}_{45}\text{Rh}$	102.91	50.00
$^{191}_{77}\text{Ir}$	190.96	15.18
$^{193}_{77}\text{Ir}$	192.96	34.82

(i) Define relative isotopic mass.

[2]

(ii) Use Table 1.2 to calculate the relative atomic mass, A_r , of iridium in the alloy.

Give your answer to **two** decimal places.

relative atomic mass of iridium = [2]

(d) Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$.

Show your working.

$x =$

[2]

17. M/J 24/P21/Q2/d(i)

(d) Boron nitride is a white solid that melts above 2900 °C.

Fig. 2.2 shows part of the lattice structure of a crystal of boron nitride.

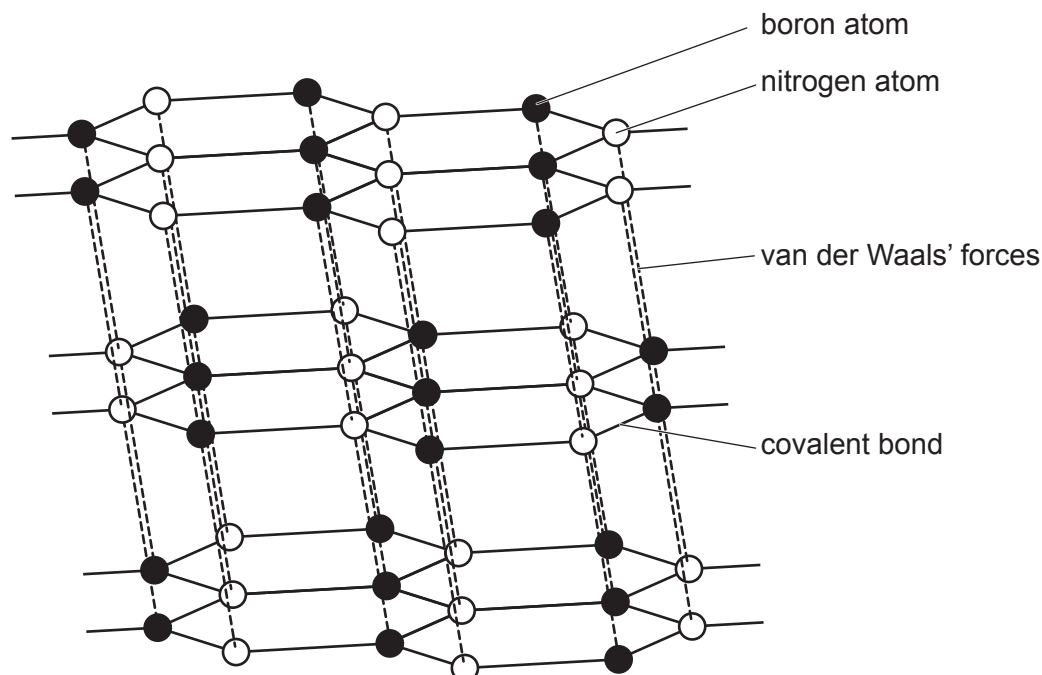


Fig. 2.2

(i) Use Fig. 2.2 to deduce the empirical formula of boron nitride.

[1]

18. M/J 24/P23/Q3/c(ii)

(c) Boron carbide is a hard crystalline solid that has a melting point greater than 2000 °C.

(ii) 100 g of pure boron carbide contains 78.26 g of boron.

Calculate the empirical formula of boron carbide.

Show your working.

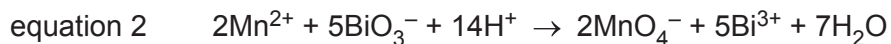
empirical formula of boron carbide

[2]

19. MAR 24/P22/Q1/e

1 Bismuth is an element in Group 15 of the Periodic Table.

- (e) NaBiO_3 can be used to determine the concentration of $\text{Mn}^{2+}(\text{aq})$. The ionic equation for the reaction is shown in equation 2.



A student uses the following procedure in an experiment.

- Add 100.0 cm^3 of a saturated solution of $\text{Mn}^{2+}(\text{aq})$ to a volumetric flask.
- Add distilled water to the flask to make a 1.00 dm^3 diluted solution.
- Titrate a 25.00 cm^3 sample of the diluted solution with 0.100 mol dm^{-3} $\text{NaBiO}_3(\text{aq})$.

The 25.00 cm^3 sample of the diluted solution of $\text{Mn}^{2+}(\text{aq})$ reacts completely with exactly 21.50 cm^3 of 0.100 mol dm^{-3} $\text{NaBiO}_3(\text{aq})$.

Calculate the concentration, in mol dm^{-3} , of $\text{Mn}^{2+}(\text{aq})$ in the saturated solution.

Show your working.

concentration of $\text{Mn}^{2+}(\text{aq})$ in the saturated solution = mol dm^{-3} [3]

20. M/J 23/P21/Q2/b(i)

2 A neutralisation reaction occurs when $\text{NaOH}(\text{aq})$ is added to $\text{H}_2\text{SO}_4(\text{aq})$.



- (b) An experiment is carried out to calculate ΔH_{neut} for the reaction between $\text{NaOH}(\text{aq})$ and $\text{H}_2\text{SO}_4(\text{aq})$.

100 cm^3 of 1.00 mol dm^{-3} $\text{NaOH}(\text{aq})$ is added to 75 cm^3 of 1.00 mol dm^{-3} $\text{H}_2\text{SO}_4(\text{aq})$ in a polystyrene cup and stirred. Results from the experiment are shown in Table 2.1.

Table 2.1

initial temperature of $\text{NaOH}(\text{aq})/^{\circ}\text{C}$	20.0
initial temperature of $\text{H}_2\text{SO}_4(\text{aq})/^{\circ}\text{C}$	20.0
maximum temperature of mixture/ $^{\circ}\text{C}$	27.8

- (i) Use equation 1 to calculate the amount, in mol, of $\text{H}_2\text{SO}_4(\text{aq})$ that is neutralised in the experiment.

amount of $\text{H}_2\text{SO}_4(\text{aq})$ neutralised = mol [1]

21. M/J 23/P21/Q4/b(ii)

4 **V** is a colourless liquid.

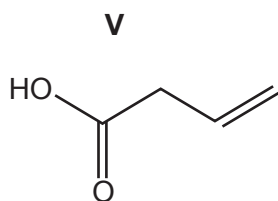


Fig. 4.1

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.

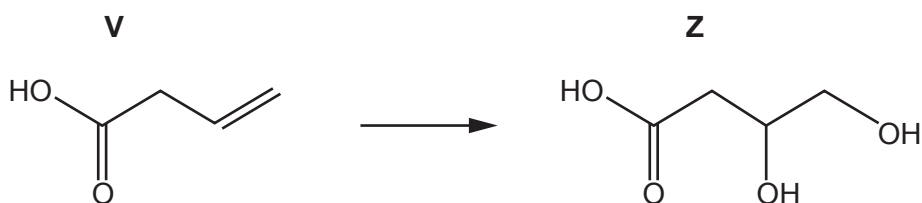


Fig. 4.2

(ii) Deduce the empirical formula of **Z**.

[1]

22. M/J 23/P22/Q2/e(iii)

2 Chlorine is a reactive element. It forms many compounds.

(e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaClO , to water.

(iii) A sample of bleach **W** contains an unknown concentration of sodium chlorate(I).

10.0 cm^3 of **W** is diluted with distilled water to make a total volume of 100 cm^3 of bleach solution. 25.0 cm^3 of this diluted bleach solution is added to an excess of hydrogen peroxide and the volume of gas produced measured under room conditions. The experiment is repeated and on average 25.0 cm^3 of diluted bleach solution produces 42.0 cm^3 of gas.

Calculate the concentration, in g dm^{-3} , of sodium chlorate(I) in **W**.

concentration of NaClO in **W** = g dm^{-3} [3]

23. M/J 23/P23/Q1/b

- 1 Copper is used in electrical equipment. It has a melting point of 1085 °C.
- (b) The relative isotopic masses and natural abundances of the two isotopes in a sample of copper are shown in Table 1.1.

Table 1.1

isotope	relative isotopic mass	% abundance
^{63}Cu	62.930	69.15
^{65}Cu	64.928	30.85

- (i) Define the unified atomic mass unit.

[1]

- (ii) Define relative atomic mass, A_r , in terms of the unified atomic mass unit.

[1]

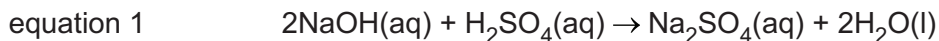
- (iii) Calculate the relative atomic mass, A_r , of copper in this sample using the data in Table 1.1.

Show your working.

$A_r =$ [1]

24. M/J 23/P23/Q3/b(i)

- 3 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



- (b) In an experiment, 50.0 cm³ of 2.00 mol dm⁻³ NaOH(aq) is added to 60.0 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Both solutions have a temperature of 21.4 °C before mixing. The maximum temperature of the mixture is measured.
- (i) Use equation 1 to calculate the amount, in mol, of H₂SO₄(aq) that is neutralised in the experiment.

amount of H₂SO₄ neutralised = mol [1]

25. M/J 23/P23/Q5/a

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

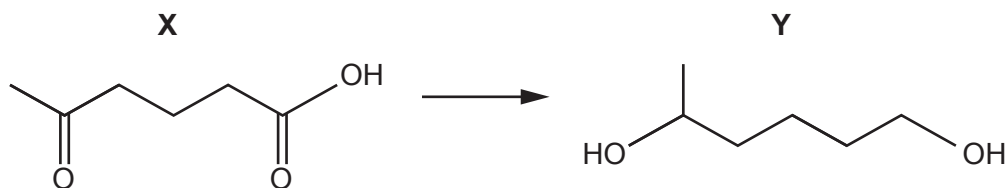


Fig. 5.1

(a) Deduce the empirical formula of **Y**.

[1]

26. MAR 23/P22/Q2/d(ii)

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(ii) Compound **X**, an anhydrous Group 2 bromide, is dissolved in water and titrated against aqueous silver nitrate.

A solution containing 0.250 g of **X** requires 33.65 cm³ of 0.0500 mol dm⁻³ AgNO₃(aq) for complete reaction.

Identify **X**.

Show your working.

X =

[3]

27. MAR 23/P22/Q3/a(iv)

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

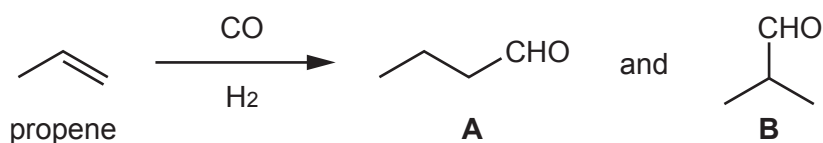


Fig. 3.1

(iv) The complete reaction of propene with a 1:1 mixture of CO and H₂ produces **A** and **B** only. The product mixture contains 96% **A** and 4% **B**.

Calculate the mass of **A** produced in this reaction when 5.00×10^3 kg of propene is used.

mass of **A** =

kg [1]

28. O/N 22/P21/Q1/a(ii,iii)

- 1 Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

(ii) Define relative atomic mass.

[2]

- (iii) The relative atomic mass of gallium, A_r , is 69.723.
The relative isotopic masses of ^{69}Ga and ^{71}Ga are:

^{69}Ga , 68.926; ^{71}Ga , 70.925.

Use this information to calculate the percentage abundance of ^{69}Ga in elemental gallium.
Show your working.

Assume that the element contains only the ^{69}Ga and ^{71}Ga isotopes.
Give your answer to **four** significant figures.

percentage abundance of ^{69}Ga = %
[2]

29. O/N 22/P22/Q1/d

- 1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(d) There are many naturally occurring hydrated compounds that contain the anion PO_4^{3-} .

(i) Name the anion PO_4^{3-} .

[1]

- (ii) Struvite is a soft hydrated mineral with $M_r = 245.3$. The anhydrous form of the mineral has the formula NH_4MgPO_4 .

Calculate the number of molecules of water of crystallisation in struvite.

Give your answer to the nearest integer. Show your working.

number of molecules of water of crystallisation = [2]

30. O/N 22/P23/Q1/a(ii,iii)

- 1 Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

(ii) Define relative atomic mass.

[2]

- (iii) The relative atomic mass of gallium, A_r , is 69.723.
The relative isotopic masses of ^{69}Ga and ^{71}Ga are:

^{69}Ga , 68.926; ^{71}Ga , 70.925.

Use this information to calculate the percentage abundance of ^{69}Ga in elemental gallium.
Show your working.

Assume that the element contains only the ^{69}Ga and ^{71}Ga isotopes.
Give your answer to **four** significant figures.

percentage abundance of ^{69}Ga = %
[2]

31. M/J 22/P21/Q1/e(i,ii)

- 1 Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

(e) A sample of magnesium contains three isotopes, ^{25}Mg , ^{26}Mg and **X**.

The percentage abundance of the three isotopes is shown in Table 1.1.

Table 1.1

isotope of Mg	mass/a.m.u.	percentage abundance/%
X		78.99
^{25}Mg	24.99	10.00
^{26}Mg	25.98	11.01

- (i) The relative atomic mass, A_r , is calculated by comparing the average mass of the isotopes of an element to the unified atomic mass unit.

Define the unified atomic mass unit.

[1]

- (ii) Calculate the mass of **X**. Use data from Table 1.1 and A_r (magnesium) = 24.31 in your calculation. Show your working.

mass of **X** =
[2]

32. M/J 22/P21/Q4/a,b(ii)

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

(a) Calculate the molecular formula of **V**. Show your working.

molecular formula of **V** = [3]

(b) **V** contains two types of functional group: a carboxylic acid and an alkene.

(ii) A 3.196 g sample of Br_2 reacts completely with 2.800 g of **V**.

Calculate how many alkene functional groups are present in one molecule of **V**. Show your working.

number of alkene functional groups in **V** = [1]

33. M/J 22/P22/Q6/b

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

(b) Table 6.2 shows the percentage by mass of each element present in **Z**.

Table 6.2

element	percentage by mass/%
carbon	41.38
hydrogen	3.45
oxygen	55.17

Using the data in Table 6.2, demonstrate that the empirical formula of **Z** is CHO . Show your working.

[1]

34. M/J 22/P23/Q1/f(i)

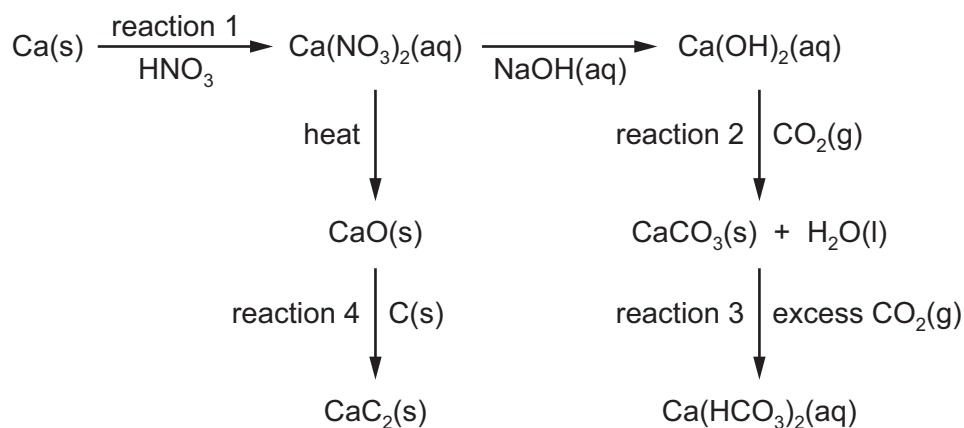
(f) (i) The element copper has a relative atomic mass of 63.5.

Calculate how many atoms are present in 1.05 g of copper.

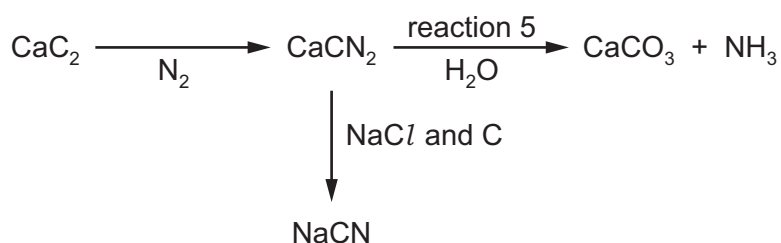
atoms of copper present = [1]

35. O/N 21/P21/Q2/d(i)

2 The reaction scheme shows some reactions of calcium.



(d) The flowchart shows some reactions of CaC_2 .



(i) Reaction 5 can be used to prepare NH_3 .



Calculate the minimum mass, in tonnes, of calcium cyanamide, CaCN_2 , that is required to produce 1.50×10^6 tonnes of NH_3 .

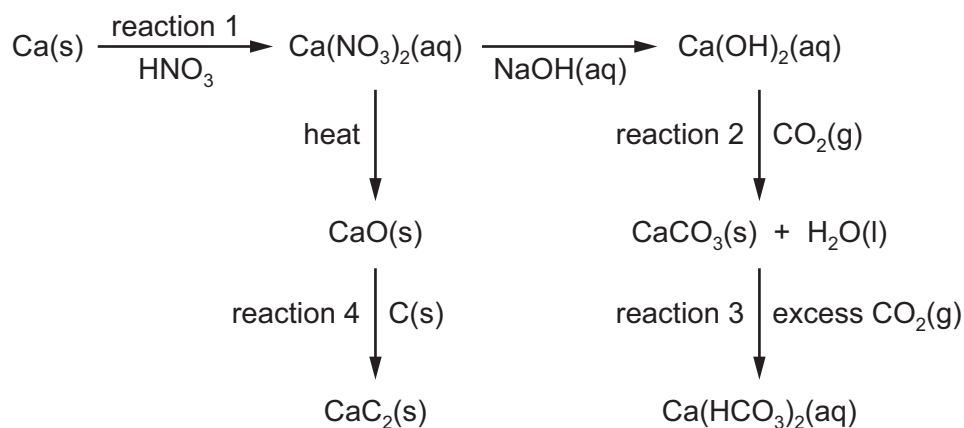
Show your working.

$$1 \text{ tonne} = 1.00 \times 10^6 \text{ g}$$

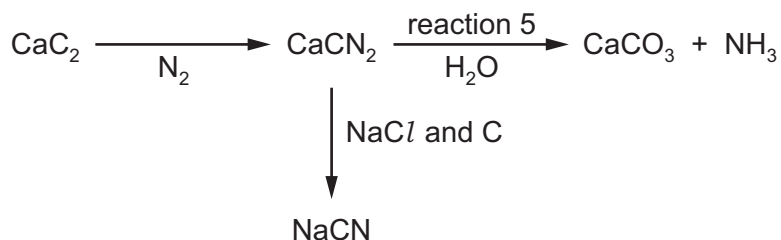
minimum mass of CaCN_2 = tonnes
[2]

36. O/N 21/P23/Q2/d(i)

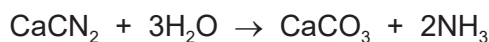
2 The reaction scheme shows some reactions of calcium.



(d) The flowchart shows some reactions of CaC_2 .



(i) Reaction 5 can be used to prepare NH_3 .



Calculate the minimum mass, in tonnes, of calcium cyanamide, CaCN_2 , that is required to produce 1.50×10^6 tonnes of NH_3 .

Show your working.

$$1 \text{ tonne} = 1.00 \times 10^6 \text{ g}$$

minimum mass of CaCN_2 = tonnes [2]

37. M/J 21/P21/Q1/a,b(i)

1 Ethanedioic acid, $\text{HO}_2\text{CCO}_2\text{H}$, has a relative molecular mass of 90.0.

(a) (i) Explain what is meant by the term *relative molecular mass*.

[2]

(ii) State the empirical formula of ethanedioic acid.

[1]

(iii) Calculate how many atoms of carbon are present in 0.18 g of ethanedioic acid, $\text{HO}_2\text{CCO}_2\text{H}$.

Show your working.

atoms of carbon present = [3]

(b) Solid ethanedioic acid reacts with aqueous calcium ions to make a precipitate of calcium ethanedioate, CaC_2O_4 .

CaC_2O_4 breaks down when heated to form calcium oxide, carbon dioxide and carbon monoxide.

(i) Construct an equation to represent the reaction of CaC_2O_4 when heated. Include state symbols.

[2]

38. M/J 21/P22/Q1/c

1 A Group 2 metal combines with bromine to form a crystalline solid, MBr_2 .

Excess aqueous AgNO_3 is added to a solution of MBr_2 and a precipitate forms. The mixture is filtered. The precipitate is dried and the mass of the precipitate is recorded.

(c) A 0.250 g sample of pure MBr_2 contains 8.415×10^{-4} mol MBr_2 .

Calculate the relative formula mass, M_r , of MBr_2 . Use this to identify **M**.

Show your working.

$M_r =$

M =

[3]

39. M/J 21/P23/Q1/e

1 The elements in Group 17 are known as the halogens.

(e) The concentration of NaClO in bleach **S** is $x \text{ g dm}^{-3}$.

NaClO reacts with $\text{H}_2\text{O}_2(\text{aq})$ as shown.



A 5.00 cm^3 sample of **S** completely reacts with $\text{H}_2\text{O}_2(\text{aq})$. The volume of $\text{O}_2(\text{g})$ produced is 24.0 cm^3 under room conditions.

Assume that only the NaClO in **S** reacts with $\text{H}_2\text{O}_2(\text{aq})$.

Calculate x . Show your working.

$x =$ g dm^{-3}
[3]

40. M/J 21/P23/Q3/c

3 Separate samples of **R**, **S**, **T** and **U** are added to cold water. The identity of each sample is unknown. However, each sample is known to be pure and can only be one of $\text{Ba}(\text{OH})_2$, NaCl, P_4O_{10} or SiCl_4 .

(c) Tungsten oxide, W_xO_y , is used to give colour to ceramic materials.

A sample of W_xO_y contains 79.29% tungsten by mass.

Calculate the empirical formula of W_xO_y .

Show your working.

empirical formula = [3]

41. MAR 21/P22/Q2/f(ii)

2 Chlorine, Cl_2 , is a reactive yellow-green gas. It is a strong oxidising agent.

(f) 0.020 mol of element **Z** reacts with excess Cl_2 to form 0.020 mol of a liquid chloride.

The liquid chloride has formula ZCl_n , where n is an integer.

ZCl_n reacts vigorously with water at room temperature to give an acidic solution and a white solid.

When excess $\text{AgNO}_3(\text{aq})$ is added to the solution, 11.54 g of $\text{AgCl}(\text{s})$ forms.

(ii) Calculate the value of n in ZCl_n .

$n =$ [2]

42. M/J 20/P21/Q1/a(ii)

1 Gallium is a metal in Group 13 of the Periodic Table.

(a) There are two stable isotopes of gallium, ^{69}Ga and ^{71}Ga .

(ii) State what further information is needed to calculate the relative atomic mass of gallium.

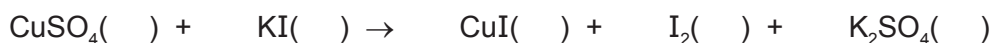
[1]

43. M/J 20/P21/Q2/b

Data Booklet needed

2 (a) The equation shown in (a)(i) describes the reaction which occurs when aqueous potassium iodide is added to aqueous copper(II) sulfate. A white precipitate of copper(I) iodide forms in a brown solution of iodine and potassium sulfate.

(i) Balance the equation and include state symbols.



[2]

The table gives the oxidation numbers of iodine in the different species in the equation.

iodine-containing species	oxidation number of iodine
KI	-1
CuI	-1
I_2	0

(b) In the reaction described in (a)(i), a student uses 17.43 g of $\text{CuSO}_4 \cdot y\text{H}_2\text{O}$. By further titration of the reaction products the student concludes that the total amount of CuSO_4 in the sample is 0.0982 mol.

Use the *Data Booklet* to complete the table to calculate the value of **y**, where **y** is an integer. Show your working.

mass of 0.0982 mol CuSO_4	
amount of H_2O in 17.43 g of $\text{CuSO}_4 \cdot y\text{H}_2\text{O}$	
value of y	

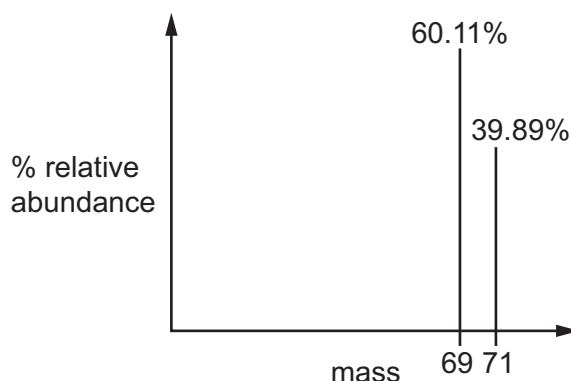
y =

[4]

44. M/J 20/P22/Q1/a,b

- 1 Gallium is an element in Group 13.

A sample of gallium is analysed using a mass spectrometer. The mass spectrum produced is shown.



- (a) Explain what is meant by the term *relative atomic mass*.

[2]

- (b) Calculate the relative atomic mass of gallium in this sample. Give your answer to 4 significant figures.

Show your working.

relative atomic mass =

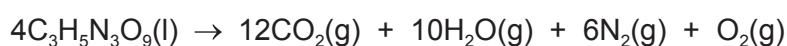
[2]

45. MAR 20/P22/Q3/b(i)

- 3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

- (b) Glycerol can be used as a starting material in the manufacture of nitroglycerine, $\text{C}_3\text{H}_5\text{N}_3\text{O}_9$.

Nitroglycerine decomposes rapidly on heating to form a mixture of gases.



A sample of nitroglycerine decomposes, releasing 1.06 dm^3 of $\text{O}_2(\text{g})$ at 850 K and $1.00 \times 10^5\text{ Pa}$.

- (i) Calculate the mass of nitroglycerine that decomposes.

mass of nitroglycerine =

g [3]

46. O/N 19/P21/Q3/b(i)

- 3 Crude oil is a natural source of hydrocarbons that are used as fuels.

- (b) Thiophene, $\text{C}_4\text{H}_4\text{S}(\text{l})$, is an organic compound that is found as a contaminant in crude oil.

- (i) Construct the equation for the complete combustion of thiophene, $\text{C}_4\text{H}_4\text{S}(\text{l})$.

Include state symbols in your answer.

[2]

47. O/N 19/P22/Q1/b(i),c(i)

- 1 In the Periodic Table, the p block contains elements whose outer electrons are found in the p subshell.
- (b) Silicon is found in many compounds in the Earth's crust. Silicon has only three naturally occurring isotopes, ^{28}Si , ^{29}Si and ^{30}Si .
- (i) The table shows data for ^{28}Si , ^{29}Si and ^{30}Si .

	^{28}Si	^{29}Si	^{30}Si
relative isotopic mass	28.0	29.0	30.0

A sample of silicon contains 92.2% ^{28}Si . The total percentage abundance of ^{29}Si and ^{30}Si in the sample is 7.8%.

The relative atomic mass, A_r , of silicon in the sample is 28.09.

Calculate the percentage abundance of ^{30}Si .

Give your answer to **one** decimal place.

percentage abundance of ^{30}Si = %
[3]

- (c) Sulfur-containing compounds, such as $\text{C}_2\text{H}_5\text{SH}$, are found in fossil fuels, and produce SO_2 when they are burned.

- (i) Write the equation to show the complete combustion of $\text{C}_2\text{H}_5\text{SH}$.

[1]

48. O/N 19/P23/Q3/b(i)

- 3 Crude oil is a natural source of hydrocarbons that are used as fuels.

- (b) Thiophene, $\text{C}_4\text{H}_4\text{S}(\text{l})$, is an organic compound that is found as a contaminant in crude oil.

- (i) Construct the equation for the complete combustion of thiophene, $\text{C}_4\text{H}_4\text{S}(\text{l})$.

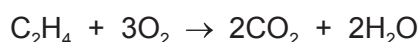
Include state symbols in your answer.

[2]

49. M/J 19/P22/Q4/c

- 4 There are many different types of aliphatic and aromatic hydrocarbons.

- (c) The equation for the complete combustion of ethene is shown.



Calculate the volume, in dm^3 , of carbon dioxide formed in the complete combustion of 1.00 g of ethene at room temperature and pressure.

volume of CO_2 = dm^3 [3]

50. M/J 19/P22/Q5/b

5 Many naturally occurring esters are used as flavourings in food.

(b) Ester **W** is made up of 54.5% carbon, 9.1% hydrogen and 36.4% oxygen.

(i) Calculate the empirical formula of **W**.

[3]

(ii) State what additional information is required to determine the molecular formula of **W**.

[1]

51. M/J 19/P23/Q1/b

(b) A sample of sulfur contains only two isotopes, ^{32}S and ^{34}S . The relative atomic mass of this sample is 32.09.

isotope	isotopic mass
^{32}S	32.0
^{34}S	34.0

Calculate the percentage abundance of the isotopes present in this sample.

% abundance ^{32}S =

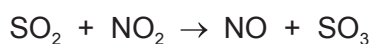
% abundance ^{34}S =

[3]

52. M/J 19/P23/Q4/b(i)

4 Release of sulfur dioxide, SO_2 , into the atmosphere causes acid rain.

(b) The SO_2 released can be converted in the atmosphere into sulfuric acid, H_2SO_4 , by reaction with nitrogen dioxide gas, NO_2 , and water, H_2O .



During one year, 1590 tonnes of SO_2 was released into the atmosphere by a fossil-fuel burning power station.

(i) Use the equations to calculate how many tonnes of H_2SO_4 were formed in the atmosphere. Assume that all of the SO_2 released was converted into H_2SO_4 .

mass of H_2SO_4 = tonnes [2]

53. MAR 19/P22/Q2/a(i,iv)

- 2** The elements in Group 17 of the Periodic Table are called the halogens. They form stable compounds with both metals and non-metals.

The table gives some data about F_2 , HCl and CaF_2 .

	F_2	HCl	CaF_2
boiling point / K	85	188	2773
relative formula mass	38.0	36.5	78.1

- (a) (i)** State what is meant by the term *relative formula mass*.

[2]

- (iv)** $CaF_2(aq)$ can be made by the reaction of calcium carbonate with hydrofluoric acid, $HF(aq)$.

Write an equation for this reaction. Include state symbols.

[2]

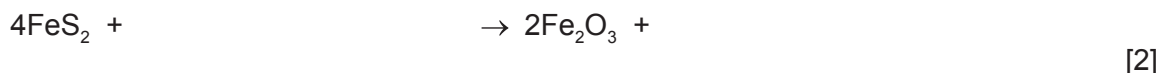
54. O/N 18/P21/Q1/c(ii,iii,iv)

- 1 Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

- (c) Iron pyrite is often called *fool's gold* because of its appearance. Impure samples of iron pyrite often contain a small amount of gold.

The gold can be obtained from impure iron pyrite. The impure iron pyrite is roasted in oxygen, to produce iron(III) oxide and sulfur dioxide. Gold does not react with oxygen.

- (ii) Complete the equation to show the roasting of iron pyrite in oxygen.



- (iii) A sample of impure iron pyrite was roasted in oxygen. The composition of the mixture of solid products is shown.

solid product	mass/g
Fe ₂ O ₃	33.18
Au	0.37

Calculate the mass of FeS_2 present in the sample of impure iron pyrite.
Assume that all the FeS_2 was converted to Fe_2O_3 during the roasting process.

(M_r : FeS₂, 120.0; Fe₂O₃, 159.6)

mass of FeS_2 = g [2]

- (iv)** Use your answer to **(iii)** to calculate the percentage by mass of gold in this sample of impure iron pyrite. Assume that gold is the only impurity in this sample of impure iron pyrite.

Give your answer to **two** significant figures.

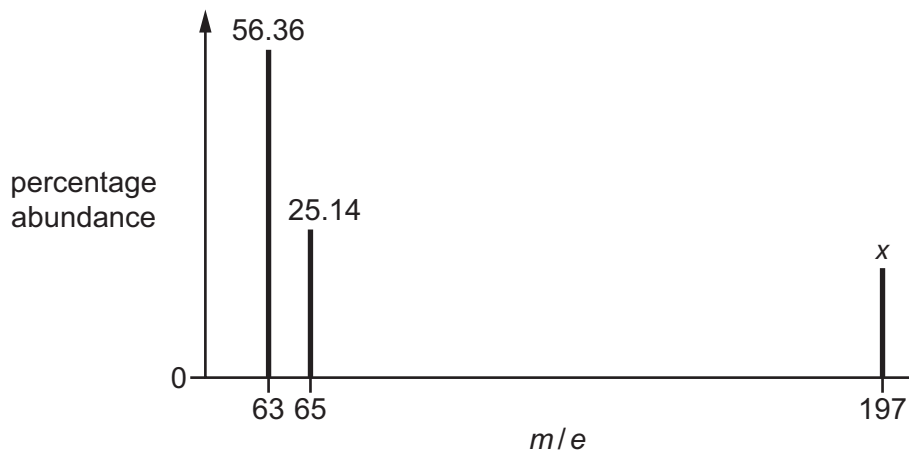
(If you were unable to calculate an answer to(iii), use 55.00 g as the mass of FeS₂ in this calculation. This is **not** the correct answer.)

percentage by mass of gold = % [1]

55. O/N 18/P22/Q1/d

- 1 The model of the nuclear atom was first proposed by Ernest Rutherford. He developed this model on the basis of results obtained from an experiment using gold metal foil.

(d) *Tumbaga* is an alloy of copper and gold. A sample of tumbaga was analysed. The mass spectrum of the sample is shown.



- (i) Calculate the percentage abundance of gold, x, in the sample of tumbaga.

x = % [1]

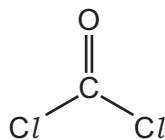
- (ii) Calculate the relative atomic mass, A_r , of the copper present in this sample. Give your answer to **two** decimal places.

A_r (Cu) = [2]

56. O/N 18/P22/Q3/c(i)

- 3 Trihalomethanes are organic molecules in which three of the hydrogen atoms of methane are replaced by halogen atoms, for example CHF_3 .

(c) A different trihalomethane, CHCl_3 , reacts with O_2 to produce carbonyl dichloride. HCl(g) is also released as a product of this reaction.



carbonyl dichloride

- (i) Write an equation for this reaction of CHCl_3 with O_2 .

[1]

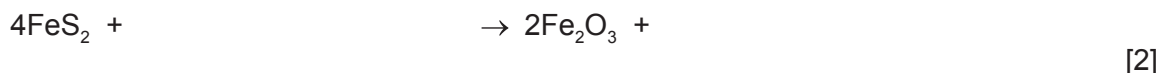
57. O/N 18/P23/Q1/c(ii,iii,iv)

- 1 Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

- (c) Iron pyrite is often called *fool's gold* because of its appearance. Impure samples of iron pyrite often contain a small amount of gold.

The gold can be obtained from impure iron pyrite. The impure iron pyrite is roasted in oxygen, to produce iron(III) oxide and sulfur dioxide. Gold does not react with oxygen.

- (ii) Complete the equation to show the roasting of iron pyrite in oxygen.



- (iii) A sample of impure iron pyrite was roasted in oxygen. The composition of the mixture of solid products is shown.

solid product	mass/g
Fe ₂ O ₃	33.18
Au	0.37

Calculate the mass of FeS_2 present in the sample of impure iron pyrite.
Assume that all the FeS_2 was converted to Fe_2O_3 during the roasting process.

(M_r : FeS₂, 120.0; Fe₂O₃, 159.6)

mass of FeS_2 = g [2]

- (iv)** Use your answer to **(iii)** to calculate the percentage by mass of gold in this sample of impure iron pyrite. Assume that gold is the only impurity in this sample of impure iron pyrite.

Give your answer to **two** significant figures.

(If you were unable to calculate an answer to (iii), use 55.00 g as the mass of FeS_2 in this calculation. This is **not** the correct answer.)

percentage by mass of gold = % [1]

58. M/J 18/P22/Q2/a,d(iii,iv,v,vi)

2 Ammonium iron(II) sulfate, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2$, has a relative formula mass, M_r , of 284.

(a) Define the term *relative formula mass*

[2]

(d) Mohr's salt, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot x\text{H}_2\text{O}$, is the hydrated form of ammonium iron(II) sulfate.

x represents the number of moles of water in 1 mole of the salt.

A student wanted to determine the value of x. 0.784 g of the hydrated salt was dissolved in water and this solution was acidified.

All of the solution was titrated with $0.0200 \text{ mol dm}^{-3}$ potassium manganate(VII). 20.0 cm^3 of this potassium manganate(VII) solution was required for complete reaction with the Fe^{2+} ions.

(iii) Calculate the amount, in moles, of manganate(VII) ions that reacted.

amount = mol [1]

(iv) Calculate the amount, in moles, of Fe^{2+} ions in the sample of the salt.

amount = mol [1]

(v) Calculate the relative formula mass of $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot x\text{H}_2\text{O}$.

relative formula mass = [1]

(vi) Calculate the value of x.

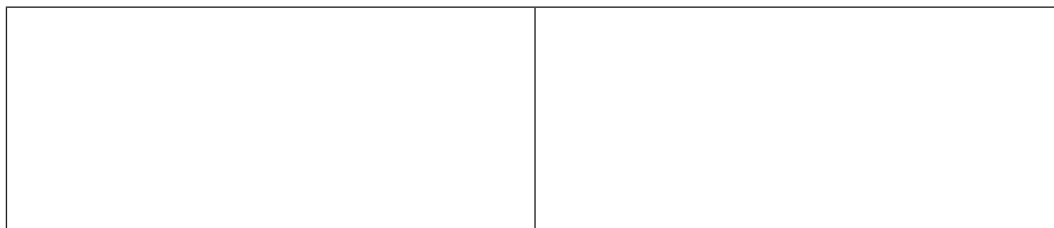
x = [1]

59. M/J 18/P22/Q4/e

4 **W** is $\text{CH}_3\text{COCH}_2\text{CH}_3$.

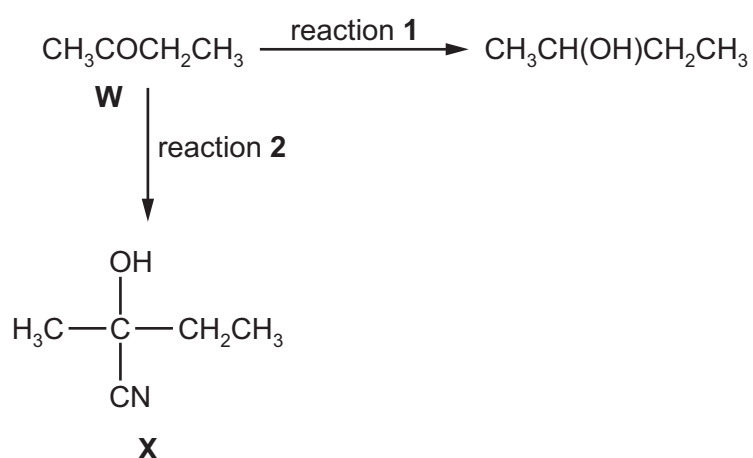
(b) There are two structural isomers of **W** that are also carbonyl compounds.

Draw the structures of these two isomers of **W**.



[2]

Two reactions of **W** are shown.



(e) If **X** is treated with ammonia and the product hydrolysed, a compound, **Y**, is obtained that contains 51.3% C, 9.40% H, 12.0% N and 27.3% O by mass.

(i) Show that the empirical formula of **Y** is $\text{C}_5\text{H}_{11}\text{NO}_2$.

[2]

(ii) The empirical formula of **Y** is $\text{C}_5\text{H}_{11}\text{NO}_2$ and the M_r of **Y** is 117.

Deduce the molecular formula of **Y**. You **must** explain your reasoning.

molecular formula =

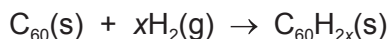
[1]

60. MAR 18/P22/Q2/c

also Unit 4 (2 marks), Unit 14 (1 mark); this unit 3 marks

2 Carbon and silicon are elements in Group 14.**(c)** 0.144 g of C_{60} was placed in a 100 cm^3 container of hydrogen gas at 20°C and $1.00 \times 10^5\text{ Pa}$.The container was heated to make the C_{60} and hydrogen gas react.

The reaction occurred as shown in the equation.

After the reaction, the container was allowed to cool to 20°C . The pressure decreased to $2.21 \times 10^4\text{ Pa}$. All of the C_{60} had reacted.**(i)** Name the type of reaction that occurred.

[1]

(ii) Calculate the amount, in moles, of C_{60} that reacted.amount of C_{60} = mol [1]**(iii)** Calculate the amount, in moles, of hydrogen gas that reacted with the C_{60} .

amount of hydrogen gas = mol [2]

(iv) Use your answers from **(ii)** and **(iii)** to deduce the molecular formula of the hydrocarbon, $C_{60}H_{2x}$.If you were unable to calculate the amount of hydrogen gas, assume that 0.00240 mol of hydrogen gas reacted. This is **not** the correct value for the amount of hydrogen gas that reacted.

molecular formula = [2]

61. M/J 17/P21/Q1/a,b(i,ii,iii)

- 1 Combustion data can be used to calculate the empirical formula, molecular formula and relative molecular mass of many organic compounds.

(a) Define the term *relative molecular mass*

[2]

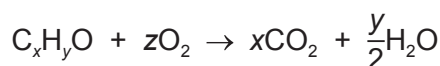
- (b) **T** is an alcohol, C_xH_yO . A gaseous sample of **T** occupied a volume of 20 cm^3 at 120°C and 100 kPa .

The sample was completely burned in 200 cm^3 of oxygen (an excess). The final volume, measured under the same conditions as the gaseous sample, was 250 cm^3 .

Under these conditions, all water present is vaporised. Removal of the water vapour from the gaseous mixture decreased the volume to 170 cm^3 .

Treating the remaining gaseous mixture with concentrated alkali, to absorb carbon dioxide, decreased the volume to 110 cm^3 .

The equation for the complete combustion of **T** can be represented as shown.



- (i) Use the data given to calculate the value of x .

$x =$ [1]

- (ii) Use the data given to calculate the value of y .

$y =$ [1]

If you were unable to calculate values for x and y then use $x = 4$ and $y = 10$ for the remaining parts of this question. These are **not** the correct values.

- (iii) Complete the equation for the complete combustion of the alcohol, **T**.



[1]

62. M/J 17/P22/Q1/b

- 1 The composition of atoms and ions can be determined from knowledge of atomic number, nucleon number and charge.
- (b) Boron occurs naturally as a mixture of two stable isotopes, ^{10}B and ^{11}B . The relative isotopic masses and percentage abundances are shown.

isotope	relative isotopic mass	abundance / %
^{10}B	10.0129	19.78
^{11}B	to be calculated	80.22

- (i) Define the term *relative isotopic mass*

[2]

- (ii) Calculate the relative isotopic mass of ^{11}B .

Give your answer to **six** significant figures. Show your working.

[2]

63. M/J 17/P23/Q1/b(i,ii,iii)

- 1 Combustion data can be used to calculate the empirical formula, molecular formula and relative molecular mass of many organic compounds. Combustion data cannot distinguish between different structural isomers.

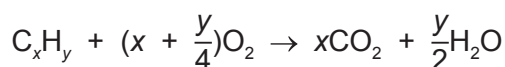
(b) **P** is a hydrocarbon, C_xH_y . A gaseous sample of **P** occupied a volume of 25 cm^3 at 37°C and 100 kPa .

The sample was completely burned in 200 cm^3 of oxygen (an excess).

The final volume, measured under the same conditions as the gaseous sample (so that the water produced is liquid and its volume can be ignored), was 150 cm^3 .

Treating the remaining gaseous mixture with concentrated alkali, to absorb carbon dioxide, decreased the volume to 50 cm^3 .

The equation for the complete combustion of **P** can be represented as shown.



- (i) Use the data given to calculate the value of x .

$$x = \quad [1]$$

- (ii) Use the data given to calculate the value of $(x + \frac{y}{4})$.

$$(x + \frac{y}{4}) = \quad [1]$$

If you were unable to calculate values in (b)(i) and (b)(ii) then use the data in this box for the remaining parts of this question. These are **not** the correct values.

$$x = 6 \qquad (x + \frac{y}{4}) = 9$$

- (iii) Give the molecular formula and the empirical formula of **P**.

molecular formula of **P**

empirical formula of **P**

[2]

64. MAR 17/P22/Q1/c(iii)

(c) Cerium is a lanthanoid metal that shows similar chemical reactions to some elements in the third period. Most of cerium's compounds contain Ce^{3+} or Ce^{4+} ions.

(iii) A naturally occurring sample of cerium contains only **four** isotopes. Data for **three** of the isotopes are shown in the table.

isotope	^{136}Ce	^{138}Ce	^{140}Ce	^{142}Ce
relative isotopic mass	135.907	137.906	139.905	to be calculated
percentage abundance	0.185	0.251	88.450	to be calculated

The A_r of the sample is 140.116.

Use these data to calculate the **relative isotopic mass** of the fourth isotope in this sample of cerium.

Give your answer to **three** decimal places.

relative isotopic mass = [3]

65. O/N 16/P21/Q1/a,b,c,d,e,f,g,h

- 1 A 0.17 g sample of a Group 14 chloride, XCl_4 , reacted with water to produce an oxide, XO_2 , and HCl .



The HCl produced was absorbed in 100 cm^3 of 0.10 mol dm^{-3} sodium hydroxide solution (an excess).

In a titration, the unreacted sodium hydroxide solution required 30.0 cm^3 of 0.20 mol dm^{-3} hydrochloric acid for complete neutralisation.

- (a) Calculate the amount, in moles, of hydrochloric acid used in the titration to neutralise the unreacted sodium hydroxide solution.

amount = mol [1]

- (b) Write the equation for the reaction between hydrochloric acid and sodium hydroxide.

[1]

- (c) Calculate the amount, in moles, of sodium hydroxide neutralised in the titration.

amount = mol [1]

- (d) Calculate the amount, in moles, of sodium hydroxide that reacted with the HCl produced by the reaction in equation 1.

amount = mol [1]

- (e) Calculate the amount, in moles, of HCl produced by the reaction in equation 1.

amount = mol [1]

- (f) Calculate the amount, in moles, of XCl_4 in the original 0.17 g sample.

amount = mol [1]

- (g) Calculate the molecular mass, M_r , of XCl_4 .

M_r = [1]

- (h) Calculate the relative atomic mass, A_r , of X and suggest its identity.

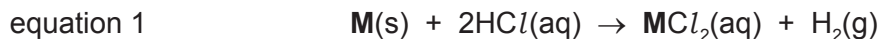
A_r of X =

identity of X

[2]

66. O/N 16/P22/Q1/a,b

- 1** A 0.50 g sample of a Group 2 metal, **M**, was added to 40.0 cm³ of 1.00 mol dm⁻³ hydrochloric acid (an excess).



- (a)** Calculate the amount, in moles, of hydrochloric acid present in 40.0 cm³ of 1.00 mol dm⁻³ HCl.

amount = mol [1]

- (b)** When the reaction had finished, the resulting solution was made up to 100 cm³ in a volumetric flask.

A 10.0 cm³ sample of the solution from the volumetric flask required 15.0 cm³ of 0.050 mol dm⁻³ sodium carbonate solution, Na₂CO₃, for complete neutralisation of the remaining hydrochloric acid.

- (i)** Write the equation for the complete reaction of sodium carbonate with hydrochloric acid.

[1]

- (ii)** Calculate the amount, in moles, of sodium carbonate needed to react with the hydrochloric acid in the 10.0 cm³ sample from the volumetric flask.

amount = mol [1]

- (iii)** Calculate the amount, in moles, of hydrochloric acid in the 10.0 cm³ sample.

amount = mol [1]

- (iv)** Calculate the total amount, in moles, of hydrochloric acid remaining after the reaction shown in equation 1.

amount = mol [1]

- (v)** Use your answers to **(a)** and **(b)(iv)** to calculate the amount, in moles, of hydrochloric acid that reacted with the 0.50 g sample of **M**.

amount = mol [1]

- (vi)** Use your answer to **(v)** and equation 1 to calculate the amount, in moles, of **M** in the 0.50 g sample.

amount = mol [1]

- (vii)** Calculate the relative atomic mass, A_r , of **M** and identify **M**.

A_r of **M** =

identity of **M** =

[2]

67. O/N 16/P23/Q1/a,b,c,d,e,f,g,h

- 1 A 0.17 g sample of a Group 14 chloride, XCl_4 , reacted with water to produce an oxide, XO_2 , and HCl .



The HCl produced was absorbed in 100 cm^3 of 0.10 mol dm^{-3} sodium hydroxide solution (an excess).

In a titration, the unreacted sodium hydroxide solution required 30.0 cm^3 of 0.20 mol dm^{-3} hydrochloric acid for complete neutralisation.

- (a) Calculate the amount, in moles, of hydrochloric acid used in the titration to neutralise the unreacted sodium hydroxide solution.

amount = mol [1]

- (b) Write the equation for the reaction between hydrochloric acid and sodium hydroxide.

[1]

- (c) Calculate the amount, in moles, of sodium hydroxide neutralised in the titration.

amount = mol [1]

- (d) Calculate the amount, in moles, of sodium hydroxide that reacted with the HCl produced by the reaction in equation 1.

amount = mol [1]

- (e) Calculate the amount, in moles, of HCl produced by the reaction in equation 1.

amount = mol [1]

- (f) Calculate the amount, in moles, of XCl_4 in the original 0.17 g sample.

amount = mol [1]

- (g) Calculate the molecular mass, M_r , of XCl_4 .

M_r = [1]

- (h) Calculate the relative atomic mass, A_r , of X and suggest its identity.

A_r of X =

identity of X

[2]

68. M/J 16/P21/Q1/d

- (d) A sample of strontium exists as a mixture of four isotopes. Information about three of these isotopes is given in the table.

mass number	86	87	88
abundance	9.86%	7.00%	82.58%

- (i) Calculate the abundance of the fourth isotope.

abundance = % [1]

- (ii) The relative atomic mass of this sample of strontium is 87.71.

Calculate the mass number of the fourth isotope.

mass number = [2]

69. M/J 16/P22/Q1/c

- (c) A sample of oxygen exists as a mixture of three isotopes. Information about two of these isotopes is given in the table.

mass number	16	17
abundance	99.76%	0.04%

- (i) Calculate the abundance of the third isotope.

abundance = % [1]

- (ii) The relative atomic mass of this sample of oxygen is 16.0044.

Calculate the mass number of the third isotope. You **must** show your working.

mass number = [2]

70. M/J 16/P22/Q3/b

- 3** Acidified potassium dichromate(VI) can oxidise ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$.
The relevant half-equations are shown.



- (b)** In an experiment a 0.242 g sample of hydrated ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4 \cdot x\text{H}_2\text{O}$, was reacted with a $0.0200 \text{ mol dm}^{-3}$ solution of acidified potassium dichromate(VI).

32.0 cm^3 of the acidified potassium dichromate(VI) solution was required for complete oxidation of the ethanedioic acid.

- (i)** Calculate the amount, in moles, of dichromate(VI) ions used to react with the sample of ethanedioic acid.

amount = mol [1]

- (ii)** Calculate the amount, in moles, of ethanedioic acid in the sample.

amount = mol [1]

- (iii)** Calculate the relative molecular mass, M_r , of the hydrated ethanedioic acid.

M_r = [1]

- (iv)** Calculate the value of x in $\text{H}_2\text{C}_2\text{O}_4 \cdot x\text{H}_2\text{O}$.

x = [1]

71. M/J 16/P23/Q1/a(i,iii,iv,v,vi,vii)

1 An experiment was carried out to determine the percentage of iron in a sample of iron wire.

(a) A 3.35 g piece of the wire was reacted with dilute sulfuric acid, in the absence of air, so that all of the iron atoms were converted to iron(II) ions. The resulting solution was made up to 250 cm³.

(i) Write a balanced equation for the reaction between the iron in the wire and the sulfuric acid.

[1]

A 25.0 cm³ sample of this solution was acidified and titrated with 0.0250 mol dm⁻³ potassium dichromate(VI). 32.0 cm³ of the potassium dichromate(VI) solution was required for complete reaction with the iron(II) ions in the sample.

The relevant half-equations are shown.



(iii) Calculate the amount, in moles, of dichromate(VI) ions used in the titration.

amount = mol [1]

(iv) Calculate the amount, in moles, of iron(II) ions in the 25.0 cm³ sample of solution.

amount = mol [1]

(v) Calculate the amount, in moles, of iron in the 3.35 g piece of wire.

amount = mol [1]

(vi) Calculate the mass of iron in the 3.35 g piece of wire.

mass = g [1]

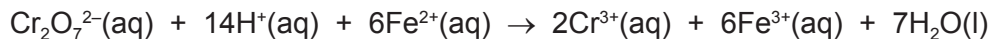
(vii) Calculate the percentage of iron in the iron wire.

percentage = % [1]

72. MAR 16/P22/Q2/a

- 2** Spathose is an iron ore that contains iron(II) carbonate, FeCO_3 . The percentage of iron(II) carbonate in spathose can be determined by titration with acidified potassium dichromate(VI) solution using a suitable indicator.

The ionic equation is shown below.



- (a)** A 5.00 g sample of spathose was reacted with excess concentrated hydrochloric acid and then filtered.

The filtrate was made up to 250 cm³ in a volumetric flask with distilled water.

A 25.0 cm³ sample of the standard solution required 27.30 cm³ of 0.0200 mol dm⁻³ dichromate(VI) solution for complete reaction.

- (i)** Calculate the amount, in moles, of dichromate(VI) ions used in the titration.

amount = _____ mol [1]

- (ii)** Use your answer to **(i)** to calculate the amount, in moles, of Fe^{2+} present in the 25.0 cm³ sample.

amount = _____ mol [1]

- (iii)** Use your answer to **(ii)** to calculate the amount, in moles, of Fe^{2+} present in the 250 cm³ volumetric flask.

amount = _____ mol [1]

- (iv)** Use your answer to **(iii)** to calculate the mass of iron(II) carbonate present in the sample of spathose.

mass = _____ g [2]

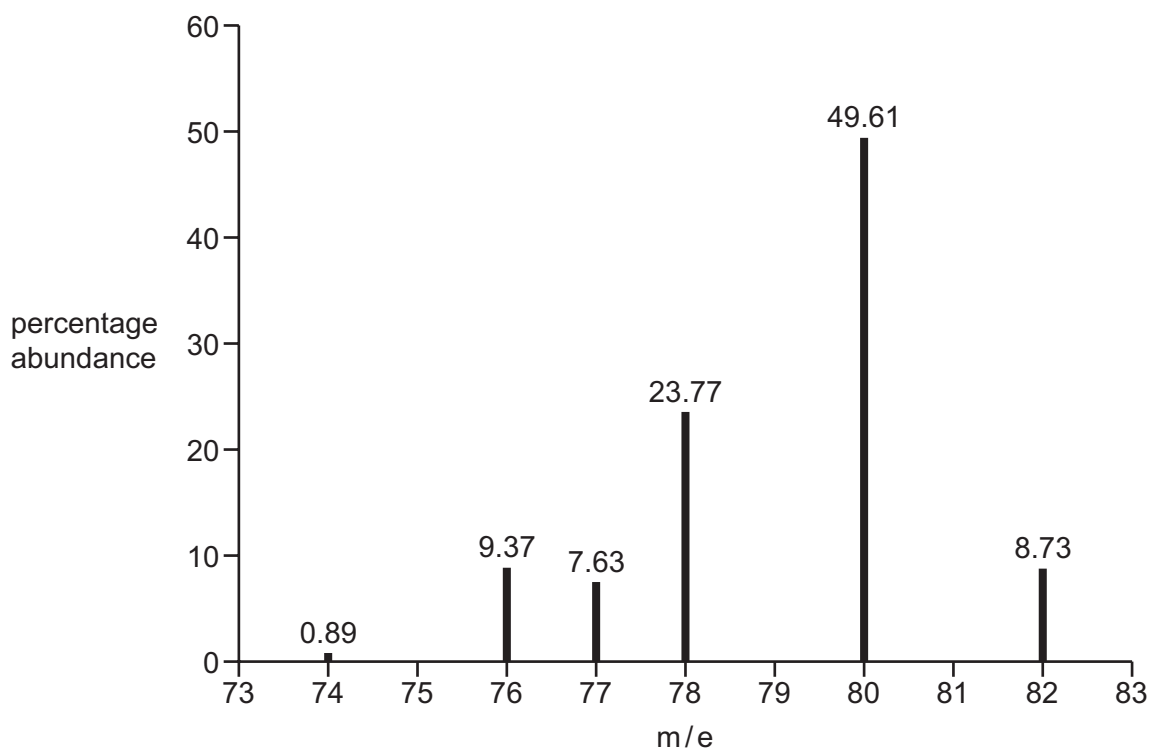
- (v)** Calculate the percentage of iron(II) carbonate in the sample of spathose.

percentage of iron(II) carbonate = _____ % [1]

73. M/J 15/P21/Q1/b(i),c(i)

- (b) The relative atomic mass of an element can be determined using data from its mass spectrum.

The mass spectrum of element **X** is shown, with the percentage abundance of each isotope labelled.



- (i) Define the terms *relative atomic mass* and *isotope*.

relative atomic mass

isotope

[3]

- (c) The element tellurium, Te, reacts with chlorine to form a single solid product, with a relative formula mass of 270. The product contains 52.6% chlorine by mass.

- (i) Calculate the molecular formula of this chloride.

molecular formula

[3]

74. M/J 15/P22/Q1/b,c(ii)

- (b) Most elements exist naturally as a mixture of isotopes, each with their own relative isotopic mass. The mass spectrum of an element reveals the abundances of these isotopes, which can be used to calculate the relative atomic mass of the element.

Magnesium has three stable isotopes. Information about two of these isotopes is given.

isotope	relative isotopic mass	percentage abundance
^{24}Mg	24.0	79.0
^{26}Mg	26.0	11.0

- (i) Define the term *relative isotopic mass*.

[2]

- (ii) The relative atomic mass of magnesium is 24.3.

Calculate the percentage abundance and hence the relative isotopic mass of the third isotope of magnesium. Give your answer to **three** significant figures

percentage abundance =

isotopic mass =

[3]

- (c) Magnesium can be produced by electrolysis of magnesium chloride in a molten mixture of salts.

- (ii) An investigation of the reaction between magnesium oxide and hydrogen chloride gas showed that an intermediate product was formed with the composition by mass Mg, 31.65%; O, 20.84%; H, 1.31% and Cl, 46.20%.

Calculate the empirical formula of this intermediate compound.

empirical formula

[2]

75. M/J 15/P22/Q2/a(i),e

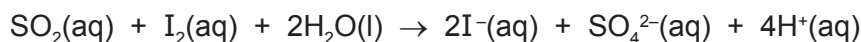
2 Sulfuric acid is an important chemical with a variety of uses.

It is manufactured by the Contact process, the first stage of which involves the conversion of sulfur or a sulfide ore, such as galena, PbS, into sulfur dioxide, SO₂.

(a) (i) Write an equation for the reaction between galena and oxygen to form sulfur dioxide and lead(II) oxide.

[2]

(e) The sulfur dioxide content of wine is most commonly measured by the Ripper Method which involves titration with iodine in the presence of starch as an indicator.



A 50.0 cm³ sample of wine required 12.35 cm³ of 0.010 mol dm⁻³ I₂(aq) for complete reaction with the SO₂.

(i) How many moles of SO₂ are present in 50.0 cm³ of wine?

moles of SO₂ in 50.0 cm³ = [1]

(ii) How many moles of SO₂ are present in 1 dm³ of wine?

moles of SO₂ in 1 dm³ = [1]

(iii) How many milligrams, mg, of SO₂ are present in 1 dm³ of wine? Give your answer to **three** significant figures. (1 g = 1000 mg)

mass of SO₂ in 1 dm³ = mg [1]

76. M/J 15/P23/Q1/c

1 Neon is a noble gas.

(c) Neon has three stable isotopes.

isotope	mass number	percentage abundance
1		9.25
2	20	90.48
3	21	0.27

(i) Define the term *relative atomic mass*.

[2]

(ii) Use the relative atomic mass of neon, 20.2, to calculate the mass number of isotope 1.

mass number = [2]

77. M/J 15/P23/Q2/d(iii)

2 The elements in Group II, and their compounds, show a variety of trends in their properties.

(d) The word 'lime' is usually used to refer to a range of calcium-containing compounds that have a range of uses.

(iii) Use your equation to calculate the mass of powdered calcium carbonate that would need to be added to the pond to neutralise the acidity.

mass = g [2]

Answers Section

1. M/J 26/P21/Q1/e

1(e)	M1 mass $Cl = 52.51$ M2 $47.49 / 32.1 : 52.51 / 35.5 (= 1.48 : 1.48)$ (empirical formula =) SCl M3 ($M_r(X) = 135.2$) molecular formula = S_2Cl_2	3 check working M1 allow $100 - 47.49$ without being evaluated A owtte $135.2 - \times(32.1) s = 2$ so $M_r = S_2Cl_2$ ✓✓✓ A alternative methods M2 A ecf from an incorrect M1 M3 A only S_2Cl_2 (with no working)
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2. M/J 26/P22/Q4/b(ii,iii)

4(b)(ii)	0.005 OR 5×10^{-3}	1	1 / 200
4(b)(iii)	1.875×10^{-3}	1	A 1.88×10^{-3} OR 1.9×10^{-3} A 2×10^{-3} with appropriate working R 1.880×10^{-3} OR 1.90×10^{-3} OR 2.0×10^{-3} OR 1.87×10^{-3}

3. M/J 26/P24/Q1/b

1(b)	$Al(NO_3)_3$	1	A $(NO_3)_3Al$
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4. O/N 25/P21/Q3/a

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	1
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5. O/N 25/P23/Q3/a

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	1
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6. O/N 25/P24/Q3/b(iv)

3(b)(iv)	$C_6H_{14}O + 9O_2 \rightarrow 6CO_2 + 7H_2O$	1
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7. M/J 25/P21/Q2/b(i)

2(b)(i)	M1 method showing each abundance multiplied by m/e value (5.9×54) and (91.9×56) and (2.2×57) OR (0.59×54) and (9.19×56) and (0.22×57) M2 the three values from M1 expressed as a percentage and added together correctly with final answer to 1 d.p $\frac{5.9}{100} \times 54 + \frac{91.9}{100} \times 56 + \frac{2.2}{100} \times 57 = 55.9$	2
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8. M/J 25/P21/Q4/a,b(i,ii)

4(a)	$C_3H_4O_3$	1
4(b)(i)	$(28.40 / 1000 \times 0.0005) = 1.42 \times 10^{-5} \text{ mol}$	1

4(b)(ii)	$(1.42 \times 10^{-5} \times 176 \times 200 / 5) = 0.099968$ M1 correct method to deduce mass of vit C in 5 cm ³ sample $(1.42 \times 10^{-5} \times 176) = 2.49 \times 10^{-3}$ M2 correct calculation for mass of vit C in original solution $(2.49 \times 10^{-3} \times 200 / 5) = 0.099968$ OR M1 correct method to deduce amount, mol, vit C in 5 cm ³ sample $(1.42 \times 10^{-5} \times 200 / 5) = 5.68 \times 10^{-4}$ M2 correct calculation for mass of vit C in original solution = M1 $\times$ 176 $(5.68 \times 10^{-4}) \times 176 = 0.099968$	2
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9. M/J 25/P22/Q2/a

2(a)(i)	route 1 M1 average (weighted) mass of (an) atom(s) (of an element) OR average (weighted) mass of the isotope(s) (of an element) M2 compared to the mass of the unified atomic mass unit OR on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 (units) OR on a scale in which the mass of a carbon-12 atom/isotope is exactly 12 (units) OR compared to $\frac{1}{12}$ mass of one atom of C-12 route 2 M1 average mass of one mol of isotopes / atoms (of an element) M2 relative / compared to $\frac{1}{12}$ of the mass of one mol of C-12 atoms/isotopes OR when the mass of one mol C-12 atoms / isotopes is exactly 12 (000) g	2
2(a)(ii)	M1 relative abundance of unknown isotope = $(100 - 91.9 - 6.0) = 2.1\%$ M2 $55.8 = (53.9 \times 6.0 + 55.9 \times 91.9 + ? \times M1) / 100 = 56.9$	2

10. M/J 25/P22/Q5/a,b(i,ii)

5(a)	C ₃ H ₄ O ₃	1
5(b)(i)	$5.00 \times 10^{-3} \times 36.65 / 1000$ $= 1.833 \times 10^{-4} / 1.8325 \times 10^{-4}$	1
5(b)(ii)	M1 expression and / or correct calculation to show how the amount of vitamin C in 150 g of lemon peel is calculated $1.8325 \times 10^{-4} \text{ mol} \times 176 \times 4$ M2 correct calculation based on $(M1 / 150) \times 100 = 0.086(\%)$ Using 1.8325×10^{-4} for M1 , M2 = 0.086 OR 0.087(%) Alternative answer using 7.65×10^{-4} : M1 $7.65 \times 10^{-4} \times 176 \times 4 = 0.5386 / 0.539$ M2 correct calculation based on $(M1 / 150) \times 100 = 0.359 / 0.36 (\%)$	2

11. M/J 25/P23/Q5/a(i)

5(a)(i)	C ₂ H ₃ O	1
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12. M/J 25/P24/Q3/a

3(a)(i)	0.00658(0) mol	1
3(a)(ii)	M1 I ₂ in 25.0 cm ³ $0.00658 / 2 = 0.00329$ (mol) M2 $0.00329 \times 1000 / 25.0 = 0.132$ (mol) alternative answer using 9.42×10^{-3} : M1 0.00471 M2 0.1884	2

13. M/J 25/P24/Q6/a(i,ii)

6(a)(i)	simplest (whole number) ratio of atoms (of the elements) in a compound	1
6(a)(ii)	C : H M1 $\frac{85.7}{12}$ or 7.14 : $\frac{14.3}{1}$ or 14.3 1 : 2 M2 CH₂	2

14. O/N 24/P21/Q1/c,d(i)

1(c)	mass of an atom of an isotope	1
	compared to the mass of the unified atomic mass unit (amu) OR on a scale on which a carbon-12 atom has a mass of exactly 12 units OR divided by $\frac{1}{12}$ mass of a carbon-12 atom	1
1(c)(ii)	$\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$	1
	= 192.35	1
1(d)(i)	M1 $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$ M2 $x = 3$	1
		1

15. O/N 24/P22/Q1/b(ii,iii)

1(b)(ii)	M1 mass of an (atom of an) isotope	1
	M2 relative/compared to (the mass of) the unified atomic mass unit OR on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 units OR divided by / compared to $\frac{1}{12}$ of the mass of a carbon-12 atom / isotope Alternative route using mass of 1 mol throughout M1 mass of one mol of an (atom of an) isotope M2 relative/compared to $\frac{1}{12}$ of the mass of 1 mol of C-12 atom / isotope OR when the mass of one mol C-12 atom / isotope is exactly 12(.000) g	1
1(b)(iii)	$92.91 \times 0.909 + 180.95 \times 0.091$	1
	= 100.92	1

16. O/N 24/P23/Q1/c,d(i)

1(c)	mass of an atom of an isotope	1
	compared to the mass of the unified atomic mass unit (amu) OR on a scale on which a carbon-12 atom has a mass of exactly 12 units OR divided by $\frac{1}{12}$ mass of a carbon-12 atom	1
1(c)(ii)	$\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$	1
	= 192.35	1
1(d)(i)	M1 $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$	1
	M2 $x = 3$	1

17. M/J 24/P21/Q2/d(i)

2(d)(i)	BN	1
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18. M/J 24/P23/Q3/c(ii)

3(c)(ii)	M1 give correct expression or calculate mol B compared to mol C $78.26 / 10.8 (= 7.246 / 7.25 \text{ mol B})$ AND $21.74 / 12 (= 1.81(2) \text{ mol C})$ M2 Find B : C ratio AND use this to state empirical formula $7.246 \text{ to } 7.25 / 1.817 \text{ to } 1.82 = 3.99 \text{ to } 3.98 \text{ mol B} : 1 \text{ mol C}$ (so) B_4C OR CB_4	2
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19. MAR 24/P22/Q1/e

1(e)	moles of $\text{NaBiO}_3 = 21.50 / 1000 \times 0.100$ ($= 2.15 \times 10^{-3} \text{ mol}$)	1
	moles of Mn^{2+} in dil. $= 2.15 \times 10^{-3} \times 2 / 5 \times 40$ ($= 0.0344 \text{ mol}$)	1
	conc. saturated $\text{Mn}^{2+} = 0.0344 \times 1000 / 100$ = 0.344 mol dm⁻³	1

20. M/J 23/P21/Q2/b(i)

2(b)(i)	0.05 mol	1
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21. M/J 23/P21/Q4/b(ii)

4(b)(ii)	CH_2O	1
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22. M/J 23/P22/Q2/e(iii)

2(e)(iii)	M1 (show how to calculate) mol O_2 gas $42 \text{ cm}^3 \text{ gas} = 42 / 24000 \text{ mol oxygen gas} (= 1.75 \times 10^{-3} \text{ mol})$ M2 (process to find) amount NaClO in 10 cm^3 sample = 100 cm^3 (diluted) bleach $\text{M1} \times 4 (= 4 \times 1.75 \times 10^{-3} \text{ mol in } 10 \text{ cm}^3 = 0.007)$ M3 correct calculation using M2 to find concentration $= (\text{M2} / 10) \times 1000 \times 74.5 = 52.2 \text{ (g / dm}^3\text{)}$	3
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23. M/J 23/P23/Q1/b

1(b)(i)	(the mass of) one twelfth of the mass of a C-12 / ^{12}C atom (or isotope)	1
1(b)(ii)	average mass of the isotopes of an element compared to the unified atomic mass unit	1
1(b)(iii)	$(62.93 \times 69.15) \div 100 + (64.928 \times 30.85) \div 100 = 63.55 / 63.546$	1

24. M/J 23/P23/Q3/b(i)

3(b)(i)	0.05 mol ($\text{H}_2\text{SO}_4(\text{aq})$)	1
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25. M/J 23/P23/Q5/a

5(a)	$\text{C}_3\text{H}_7\text{O}$	1
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26. MAR 23/P22/Q2/d(ii)

2(d)(ii)	moles of $\text{AgNO}_3 = \frac{33.65}{1000} \times 0.0500 (= 1.68(25) \times 10^{-3})$	1
	moles of $\text{X} = \frac{1}{2} \times \text{moles of AgNO}_3 = 0.250 \div M_r(\text{X}) \quad \therefore M_r(\text{X}) = 297.2$	1
	A_r of Group 2 element is $297.2 - 2(79.9) = 137.4$ AND X is BaBr_2	1

27. MAR 23/P22/Q3/a(iv)

3(a)(iv)	$96\% \times (5000 / 42.0) \times 72.0 = 8230 \text{ or } 8229 \text{ (kg)}$	1
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28. O/N 22/P21/Q1/a(ii,iii)

1(a)(ii)	M1 (weighted) average / mean mass of the isotopes / average mass of the atom(s) (of an element)	1
	M2 compared to (the mass of) the unified atomic mass unit	1
1(a)(iii)	$69.723 = 68.926x + 70.925(1-x) \therefore x = 0.6013$ $/ 69.723 = \frac{68.926x + 70.925(100-x)}{100}$	1
	60.13%	1

29. O/N 22/P22/Q1/d

1(d)(i)	phosphate(V)	1
1(d)(ii)	$(\text{molecular mass of } \text{NH}_4\text{MgPO}_4) = 137.3$	1
	$(245.3 - M_r(\text{NH}_4\text{MgPO}_4) \div 18.0) = 6$	1

30. O/N 22/P23/Q1/a(ii,iii)

1(a)(ii)	M1 (weighted) average / mean mass of the isotopes / average mass of the atom(s) (of an element)	1
	M2 compared to (the mass of) the unified atomic mass unit	1
1(a)(iii)	$69.723 = 68.926x + 70.925(1-x) \therefore x = 0.6013$ $/ 69.723 = \frac{68.926x + 70.925(100-x)}{100}$	1
	60.13%	1

31. M/J 22/P21/Q1/e(i,ii)

1(e)(i)	1 / 12 (one twelfth) the mass of a carbon-12 / ^{12}C atom	1
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1(e)(ii)	M1 correct expression relating A_r to the mass / % abundance of the three isotopes $24.31 = x \times 0.7899 + 24.99 \times 0.1000 + 25.98 \times 0.1101$ M2 correct answer to 4 sig figs atomic mass of X = 23.99	2
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32. M/J 22/P21/Q4/a,b(ii)

4(a)	M1 % / A_r for C H O M2 each % / A_r for C H O divided by the smallest value for % / A_r to give simplest whole number ratio / empirical formula M3 compare M_r from M2 ratio with 280 to deduce the actual molecular formula C 77.2 / 12 = 6.433 9(.03) H 11.4 / 1 = 11.4 16 O 11.4 / 16 = 0.7125 1 $M_r(\text{C}_9\text{H}_{16}\text{O}) = 140$ so molecular formula of V = $\text{C}_{18}\text{H}_{32}\text{O}_2$	3
4(b)(ii)	3.196 g Br_2 = 3.196 / 159.8 (= 0.02 mol Br_2) AND 2.8(00) / 280 (= 0.01 mol V) 2 alkene / 2 C=C (groups)	1

33. M/J 22/P22/Q6/b

6(b)	C 41.38 / 12 3.45 : H 3.45 / 1 3.45 : O 55.17 / 16 3.45 (so $\text{C}_{(1)}\text{H}_{(1)}\text{O}_{(1)}$)	1
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34. M/J 22/P23/Q1/f(i)

1(f)(i)	$M_r = 63.5$ $(1.05 / 63.5) \times 6.022 \times 10^{23} = 9.958 \times 10^{21}$ OR 9.96×10^{21}	1
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35. O/N 21/P21/Q2/d(i)

2(d)(i)	M1 moles of NH_3 = $1.50 \times 10^6 \times 10^6 \div 17 = 8.82 \times 10^{10}$ M2 mass of CaCN_2 = $\frac{\frac{1}{2} \times M1 \times 80.1}{10^6} = 3.53 \times 10^6$	2
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36. O/N 21/P23/Q2/d(i)

2(d)(i)	M1 moles of NH_3 = $1.50 \times 10^6 \times 10^6 \div 17 = 8.82 \times 10^{10}$ M2 mass of CaCN_2 = $\frac{\frac{1}{2} \times M1 \times 80.1}{10^6} = 3.53 \times 10^6$	2
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37. M/J 21/P21/Q1/a,b(i)

1(a)(i)	option 1 M1 the mass of a molecule OR the (weighted) average / (weighted) mean mass of the molecule(s)	1
	option 1 and M2 relative / compared to 1 / 12 (the mass) of an atom of carbon-12	1
	OR on a scale in which a carbon-12 atom / isotope has a mass of (exactly) 12 (units) option 2 M1 mass of one mol of molecules	
	option 2 M2 relative / compared to 1 / 12 (the mass) of 1 mol of C-12 OR which one mol C-12 (atom / isotope) has a mass of (exactly) 12 g	
1(a)(ii)	CO_2H	1

1(a)(iii)	0.18/90 × 2 × 6.02 × 10 ²³ = 2.408 × 10 ²¹ (atoms) OR 2.4(1) × 10²¹ (atoms) M1 no mole ethanedioic acid 0.18 / 90 = 0.0020	1
	M2 no mole ethanedioic acid × 2 0.0020 × 2 = 0.0040	1
	M3 no mole ethanedioic acid × 6.02 × 10 ²³ 2.4 × 10 ²¹	1
1(b)(i)	CaC ₂ O ₄ (s) → CaO(s) + CO ₂ (g) + CO(g) M1 correct formulae	1
	M2 balancing equation AND state symbols.	1

38. M/J 21/P22/Q1/c

1(c)	M1 calculate <i>Mr</i> of MBr₂ using 8.415 × 10 ⁻⁴ mol MBr₂ in 0.250 g <i>M_r</i> = = 297(.1).....	1
	M2 calculate the atomic mass of M using <i>M_r</i> calculated in M1 297.1 – (2 × 79.9) = 137(.4)	1
	M3 identify group 2 element from <i>A_r</i> in M2 Ba / barium	1

39. M/J 21/P23/Q1/e

1(e)	M1 Use volume O ₂ to express / find no mol O ₂ produced 24 / 24 000 mol O ₂ produced = 1 × 10 ⁻³ (mol)	1
	M2 Use 1:1 ratio to deduce no mol NaClO in 1000 cm ³ sample M1 × 1000 / 5 = 0.2 (mol dm ⁻³)	1
	M3 Use amount of NaClO and its molecular mass (74.5) to find the mass of NaClO added 24 / 24 000 × 1000 / 5 × 74.5 = 14.9 (g dm ⁻³)	1

40. M/J 21/P23/Q3/c

3(c)	% of O M1 100 – 79.29 (= 20.71)	1
	M2 express W and O as mol by / 183.8 and / 16 respectively	1
	M3 divide each by smallest number to give empirical formula WO ₃	1

41. MAR 21/P22/Q2/f(ii)

2(f)(ii)	M1: 11.54 ÷ 143.4 = 0.0805 M2: so ratio Z:Cl is 1:4 / <i>n</i> = 4	2
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42. M/J 20/P21/Q1/a(ii)

1(a)(ii)	the relative abundance / % abundance of (each) the isotopes.	1
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43. M/J 20/P21/Q2/b

2(b)	Mass of 0.0982 mol CuSO ₄ in 17.43g CuSO ₄ .yH ₂ O M1 calculate <i>M_r</i> CuSO ₄ using <i>A_r</i> from data booklet 63.5 + 32.1 + 64.0 = 159.6 M2 use <i>M_r</i> to calculate mass of CuSO ₄ (0.0982 × M1) = 15.67272g	4
	number of water in 17.43g of CuSO ₄ yH ₂ O M3 calculate the mass amount of water in sample AND use this value to calculate the amount of water present (17.43-15.67)/18 = 0.097778 mol	
	value of y M4 use the ratio of M2: 0.0982 to find y (mol H ₂ O ÷ mol CuSO ₄) = 1	
2(a)(i)	2CuSO ₄ (aq) + 4KI(aq) → 2CuI(s) + (1)I ₂ (aq) + 2K ₂ SO ₄ (aq) M1 correct balancing M2 correct state symbols	2

44. M/J 20/P22/Q1/a,b

1(a)	EITHER M1 (weighted) average/mean mass of the isotope(s)/an atom(s) M2 relative to 1/12 of the mass (of an atom) of ^{12}C (where an atom of ^{12}C is exactly 12). OR M1 mass of one mol of atoms M2 relative / compared to 1/12 (the mass) of 1 mol of C-12 OR in which one mol C-12 (atom) has a mass of (exactly) 12 g	2
1(b)	M1 $60.11/100 \times 69 + 39.89/100 \times 71$ M2 69.80	2

45. MAR 20/P22/Q3/b(i)

3(b)(i)	M1 no. of mol $\text{O}_2 = \frac{1.00 \times 10^5 \times 1.06 \times 10^{-3}}{(8.31 \times 850)}$ M2 no. of mol of nitroglycerine = $4 \times 0.0150 = 0.0600$ (mol) M3 mass of nitroglycerine = $0.0600 \times 227 = 13.6(2)$ (g)	3
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46. O/N 19/P21/Q3/b(i)

3(b)(i)	$\text{C}_4\text{H}_4\text{S(l)} + 6\text{O}_2\text{(g)} \rightarrow 4\text{CO}_2\text{(g)} + 2\text{H}_2\text{O(l)} + \text{SO}_2\text{(g)}$ - correct species - balancing - state symbols Award one mark for two correct bullet points, award two marks for all three correct.	2
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47. O/N 19/P22/Q1/b(i),c(i)

1(b)(i)	M1: $\frac{(92.2 \times 28) + ({}^{29}\text{Si} \times 29) + ({}^{30}\text{Si} \times 30)}{100} = 28.09$ M2: ($x =$) 6.6 OR $28.09 = 28.078 + x$ (where $x =$ abundance of Si-29) M3: 7.8 – M2 calculated correctly to one decimal place (or more) (=) 1.2%	3
1(c)(i)	$\text{C}_2\text{H}_5\text{SH} + 4\frac{1}{2}\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O} + \text{SO}_2$	1

48. O/N 19/P23/Q3/b(i)

3(b)(i)	$\text{C}_4\text{H}_4\text{S(l)} + 6\text{O}_2\text{(g)} \rightarrow 4\text{CO}_2\text{(g)} + 2\text{H}_2\text{O(l)} + \text{SO}_2\text{(g)}$ - correct species - balancing - state symbols Award one mark for two correct bullet points, award two marks for all three correct.	2
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49. M/J 19/P22/Q4/c

4(c)	method 1	method 2	
M1	$1 / 28 (= 0.035714)$	$1 : 88 / 28 (= 3.14286)$	1
M2	$2 \times \text{M1} (= 1 / 14 = 0.07143)$	$\text{M1} / 44 (= 0.071429)$	1
M3	$\text{M2} \times 24 = 1.7 \text{ dm}^3$	$\text{M2} \times 24 = 1.7 \text{ dm}^3$	1

50. M/J 19/P22/Q5/b

5(b)(i)	M1 divide by A_r	C	H	O	1
		54.5 / 12	9.1 / 1	36.4 / 16	
	M2 divide by smallest number	4.54 / 2.275 (= 2 OR 1.99)	9.1 / 2.275(=4)	2.275 / 2.275 (=1)	
	M3 empirical formula based on correctly rounded values of M2	C ₂ H ₄ O			1
5(b)(ii)	(relative) molecular mass / M_r				1

51. M/J 19/P23/Q1/b

1(b)	M1 $x / 100 \times 32 + (100 - x / 100 \times 34) = 32.09$ M2 $(32x + 3400 - 34x) = 3209$ so $x = 95.5$ M3 S ³² 95.5% AND S ³⁴ 4.5%	3
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52. M/J 19/P23/Q4/b(i)

4(b)(i)	M1 1 mol SO ₂ → 1 mol H ₂ SO ₄ 64.1 g / tonne → 98.1 g / tonne M2 SO ₂ → 98.1 / 64.1 × 1590 = 2433.369735 tonnes	2
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53. MAR 19/P22/Q2/a(i,iv)

2(a)(i)	M1 ① mass of a molecule OR ② (weighted) average / mean mass of the molecules OR ③ mass of one mole of molecules M2 ① / ② compared to $\frac{1}{12}$ (the mass) of an atom of carbon-12 OR on a scale in which a carbon-12 atom / isotope has a mass of (exactly) 12 (units) ③ relative / compared to $\frac{1}{12}$ (the mass) of 1 mole of carbon-12 OR on a scale in which 1 mole of carbon-12 (atoms / isotope) has a mass of (exactly) 12 g	2
2(a)(iv)	CaCO ₃ (s) + 2HF(aq) → CaF ₂ (aq) + CO ₂ (g) + H ₂ O(l) M1 species and balancing M2 state symbols	2

54. O/N 18/P21/Q1/c(ii,iii,iv)

1(c)(ii)	balanced equation with 11O ₂ and 8SO ₂ M1 : O ₂ and SO ₂ M2 : 11 and 8	2
1(c)(iii)	M1 is for process of calculating number of moles of Fe ₂ O ₃ 33.18 ÷ 159.6 (= 0.2079 mol) M2 for correct use of stoichiometry and 120.0 with candidate's M1 M2 (0.2079) × 4 / 2 × 120.0 = 49.89 (g)	2
1(c)(iv)	(0.37 / (0.37 + 49.89)) = 0.74	1

55. O/N 18/P22/Q1/d

1(d)(i)	(100 – 56.36 – 25.14) = 18.5(0)	1
1(d)(ii)	M1 correct use of ⁶³ Cu and ⁶⁵ Cu and their % abundance [1] M2 ÷ (56.36 + 25.14) AND answer correct to two decimal places [1]	2

56. O/N 18/P22/Q3/c(i)

3(c)(i)	$\text{CHCl}_3 + \frac{1}{2}\text{O}_2 \rightarrow \text{COCl}_2 + \text{HCl}$	1
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57. O/N 18/P23/Q1/c(ii,iii,iv)

1(c)(ii)	balanced equation with 11O_2 and 8SO_2 M1: O_2 and SO_2 M2: 11 and 8	2
1(c)(iii)	M1 is for process of calculating number of moles of Fe_2O_3 $33.18 \div 159.6 (= 0.2079 \text{ mol})$ M2 for correct use of stoichiometry and 120.0 with candidate's M1 M2 $(0.2079) \times 4/2 \times 120.0 = 49.89 \text{ (g)}$	2
1(c)(iv)	$(0.37/(0.37+49.89)) = 0.74$	1

58. M/J 18/P22/Q2/a,d(iii,iv,v,vi)

2(a)	<i>option 1:</i> the mass of (all the atoms/ions in) a formula (unit) / molecule OR the (weighted) average / (weighted) mean mass of (all the atoms / ions in) the formula (unit) / molecule [1] relative / compared to 1 / 12 (the mass of an atom) of carbon-12 OR on a scale in which a carbon-12 (atom / isotope) has a (mass) of (exactly) 12 (units) [1] <i>option 2:</i> mass of one mol of a compound / formula (unit) / molecule [1] relative / compared to 1 / 12 (the mass) of 1 mol of C-12 OR in which one mol C-12 (atom / isotope) is (a mass of exactly) 12 g [1]	2
2(d)(iii)	$4 \times 10^{-4} / 0.0004$	1
2(d)(iv)	$2 \times 10^{-3} / 0.002$	1
2(d)(v)	392	1
2(d)(vi)	6	1

59. M/J 18/P22/Q4/e

4(e)(i)	<i>M1 is for the process of taking the % of each element and dividing by its relative atomic mass.</i> <table><tr><td>C</td><td>H</td><td>N</td><td>O</td></tr><tr><td>$\frac{51.3}{12}$</td><td>$\frac{9.40}{1}$</td><td>$\frac{12.0}{14}$</td><td>$\frac{27.3}{16}$</td></tr></table> OR <table><tr><td>4.28</td><td>9.40</td><td>0.857</td><td>1.71</td></tr></table> <i>M2 is for dividing the smallest %/A_r into each of the remaining values to produce the correct ratio.</i> <table><tr><td>$\frac{4.28}{0.857}$</td><td>$\frac{9.40}{0.857}$</td><td>$\frac{0.857}{0.857}$</td><td>$\frac{1.71}{0.857}$</td></tr></table> OR <table><tr><td>4.9942</td><td>:</td><td>10.9685</td><td>:</td><td>1</td><td>:</td><td>1.9953</td></tr></table>	C	H	N	O	$\frac{51.3}{12}$	$\frac{9.40}{1}$	$\frac{12.0}{14}$	$\frac{27.3}{16}$	4.28	9.40	0.857	1.71	$\frac{4.28}{0.857}$	$\frac{9.40}{0.857}$	$\frac{0.857}{0.857}$	$\frac{1.71}{0.857}$	4.9942	:	10.9685	:	1	:	1.9953	2
C	H	N	O																						
$\frac{51.3}{12}$	$\frac{9.40}{1}$	$\frac{12.0}{14}$	$\frac{27.3}{16}$																						
4.28	9.40	0.857	1.71																						
$\frac{4.28}{0.857}$	$\frac{9.40}{0.857}$	$\frac{0.857}{0.857}$	$\frac{1.71}{0.857}$																						
4.9942	:	10.9685	:	1	:	1.9953																			
4(e)(ii)	C ₅ H ₁₁ NO ₂ AND because the EFM = RFM	1																							

60. MAR 18/P22/Q2/c

2(c)(i)	addition	1
2(c)(ii)	$(n_{\text{C60}} = 0.144 / 720 =) 2 \times 10^{-4}$	1

2(c)(iii)	$pV = nRT \quad \therefore \quad \Delta n = (p_1 - p_2)V / RT$ $\Delta n = (1.00 \times 10^5 - 2.21 \times 10^4) \cdot 100 \times 10^{-6} / 8.31 \times 293$ $= 0.00320$	2
2(c)(iv)	$(C_{60}:H_2) = 2.00 \times 10^{-4} : 0.00320$ or 1:16 $C_{60}H_{32}$	2

61. M/J 17/P21/Q1/a,b(i,ii,iii)

1(a)	The mass of a molecule OR the (weighted) average / (weighted) mean mass of the molecules Relative / compared to $\frac{1}{12}$ (the mass) of <u>an atom</u> of carbon-12 OR on a scale in which a carbon-12 atom / isotope has a mass of (exactly) 12 (units)	1
1(b)(i)	3	1
1(b)(ii)	8	1
1(b)(iii)	$C_3H_8O + 4\frac{1}{2}O_2 \rightarrow 3CO_2 + 4H_2O$	1

62. M/J 17/P22/Q1/b

1(b)(i)	EITHER mass of an atom / isotope relative / compared to 1/12 (the mass) of (an atom of) C-12 OR on a scale in which a C-12 (atom / isotope) has (a mass of exactly) 12 (units) OR mass of one mol (of atoms) of an isotope relative / compared to 1/12 (the mass) of 1 mol of C-12 OR on a scale in which one mol C-12 (atom / isotope) has a mass of (exactly) 12 g	2
1(b)(ii)	$\frac{(10.0129 \times 19.78) + (80.22x)}{100} = 10.8$	1
	$x = 10.9941$	1

63. M/J 17/P23/Q1/b(i,ii,iii)

1(b)(i)	4	1
1(b)(ii)	6	1
1(b)(iii)	molecular = C_4H_8 empirical = CH_2 using alternative supplied data molecular = C_6H_{12} empirical = CH_2	1

64. MAR 17/P22/Q1/c(iii)

1(c)(iii)	M1 % abundance of fourth isotope $= 100 - (0.185 + 0.251 + 88.450) = 11.114$	1
	M2 $\frac{(0.185 \times 135.907) + (0.251 \times 137.906) + (88.450 \times 139.905) + (11.114 \times \text{RIM})}{100}$ $= 140.116$ $\therefore (140.116 \times 100) - 12434.35 = 1577.246 = 11.114 \times \text{RIM}$	1
	M3 $\text{RIM} = \frac{1577.246}{11.114} = 141.915$	1

65. O/N 16/P21/Q1/a,b,c,d,e,f,g,h

1(a)	6×10^{-3} (mol)	1	1
1(b)	$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$	1	1
1(c)	6×10^{-3} (mol)	1	1
1(d)	4×10^{-3} (mol)	1	1
1(e)	4×10^{-3} (mol)	1	1
1(f)	1×10^{-3} (mol)	1	1
1(g)	170	1	1
1(h)	28(.0) Si/silicon	1 1	2

66. O/N 16/P22/Q1/a,b

1(a)	0.04 OR 4×10^{-2}	1
1(b)(i)	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$	1
1(b)(ii)	0.00075 OR 7.5×10^{-4}	1
1(b)(iii)	0.0015 OR 1.5×10^{-3}	1
1(b)(iv)	0.015 OR 1.5×10^{-2}	1
1(b)(v)	0.025 OR 2.5×10^{-2}	1
1(b)(vi)	0.0125 OR 1.25×10^{-2} OR 0.013 OR 1.3×10^{-2}	1
1(b)(vii)	40 Ca / calcium	1 1

67. O/N 16/P23/Q1/a,b,c,d,e,f,g,h

1(a)	6×10^{-3} (mol)	1	1
1(b)	$\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$	1	1
1(c)	6×10^{-3} (mol)	1	1

1(d)	4×10^{-3} (mol)	1	1
1(e)	4×10^{-3} (mol)	1	1
1(f)	1×10^{-3} (mol)	1	1
1(g)	170	1	1
1(h)	28(.0) Si/silicon	1 1	2

68. M/J 16/P21/Q1/d

(d) (i)	0.56(%)	[1]	[1]
(ii)	$\frac{(A \times 0.56) + (86 \times 9.86) + (87 \times 7.00) + (88 \times 82.58)}{100} = 87.71$ $A = 84$	[1] [1]	[2]

69. M/J 16/P22/Q1/c

(c) (i)	$(100 - 99.76 - 0.04) = 0.2$	[1]	[1]
(ii)	$\frac{0.2x + (99.76 \times 16) + (0.04 \times 17)}{100} = 16.0044$ $x = 18$	[1] [1]	[2]

70. M/J 16/P22/Q3/b

(b) (i)	$(0.02 \times 32.0/1000) = 6.40 \times 10^{-4}$	[1]	[1]
(ii)	$(6.4 \times 10^{-4} \times 3) = 1.92 \times 10^{-3}$	[1]	[1]
(iii)	$(0.242 / 1.92 \times 10^{-3}) = 126(.0)$	[1]	[1]
(iv)	$(126 - 90 = 36; 36 / 18 = 2 \text{ hence}) x = 2$	[1]	[1]

71. M/J 16/P23/Q1/a(i,iii,iv,v,vi,vii)

1 (a) (i)	$\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2$	[1]	[1]
(iii)	$(0.025 \times 32.0/1000) = 8 \times 10^{-4}$	[1]	[1]
(iv)	$(8 \times 10^{-4} \times 6) = 4.8 \times 10^{-3}$	[1]	[1]
(v)	$(4.8 \times 10^{-3} \times 250/25.0) = 4.8 \times 10^{-2}$	[1]	[1]
(vi)	$(4.8 \times 10^{-2} \times 55.8) = 2.68/2.678$	[1]	[1]
(vii)	$(2.68/3.35) = 80\%$	[1]	[1]

72. MAR 16/P22/Q2/a

2 (a) (i)	$\frac{27.30}{1000} \times 0.020 = 5.46 \times 10^{-4}$ (mol)	[1]	[1]
(ii)	$(i) \times 6 = 3.28 \times 10^{-3}$ (mol)	[1]	[1]
(iii)	$(ii) \times \frac{250}{25.00} = 3.28 \times 10^{-2}$ (mol)	[1]	[1]
(iv)	$M_r \text{ of FeCO}_3 = 55.8 + 12.0 + 3(16.0) = 115.8$ $(iii) \times M_r(\text{FeCO}_3) = 3.79 \text{ g}$	[1] [1]	[2]

(v)	$\frac{(\text{iv})}{5.00} \times 100\% = 75.9\%$	[1]	[1]
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73. M/J 15/P21/Q1/b(i),c(i)

(b) (i)	RAM = mean/average mass of the isotopes/an atom(s) relative to 1/12 the mass of an atom of ^{12}C /on a scale where an atom of ^{12}C is (exactly) 12 (units)	[1] [1]	
	isotope = atoms with the same number of protons/atomic number/proton number with different mass numbers/numbers of neutrons/nucleon number	[1]	[3]

(c) (i)	<table><tr><td>Te</td><td>Cl</td><td></td></tr><tr><td>$\frac{47.4}{128}$</td><td>$\frac{52.6}{35.5}$</td><td></td></tr><tr><td>$\frac{0.370}{0.370}$</td><td>$\frac{1.48}{0.370}$</td><td></td></tr><tr><td>1</td><td>4</td><td>so EF = TeCl_4</td></tr></table>	Te	Cl		$\frac{47.4}{128}$	$\frac{52.6}{35.5}$		$\frac{0.370}{0.370}$	$\frac{1.48}{0.370}$		1	4	so EF = TeCl_4	[1]	
Te	Cl														
$\frac{47.4}{128}$	$\frac{52.6}{35.5}$														
$\frac{0.370}{0.370}$	$\frac{1.48}{0.370}$														
1	4	so EF = TeCl_4													
	Empirical Formula Mass = 270	[1]													
	so MF = TeCl_4	[1]	[3]												

74. M/J 15/P22/Q1/b,c(ii)

(b) (i)	Mass of an atom(s)	[1]																
	relative to $1/12^{\text{th}}$ (the mass) of (an atom of) carbon-12 OR relative to carbon-12 which is (exactly) 12	[1]	[2]															
(ii)	% of third isotope = 10	[1]																
	$\frac{(24 \times 79) + (26 \times 11.0) + 10x}{100} = 24.3$	[1]																
	10x = 248																	
	x = 24.8 (3s.f.)	[1]	[3]															
(ii)	<table><tr><td>Mg</td><td>O</td><td>H</td><td>Cl</td><td></td></tr><tr><td>$\frac{31.65}{24.3}$</td><td>$\frac{20.84}{16}$</td><td>$\frac{1.31}{1}$</td><td>$\frac{46.2}{35.5}$</td><td></td></tr><tr><td>1.30</td><td>1.30</td><td>1.31</td><td>1.30</td><td>= 1:1:1:1</td></tr></table>	Mg	O	H	Cl		$\frac{31.65}{24.3}$	$\frac{20.84}{16}$	$\frac{1.31}{1}$	$\frac{46.2}{35.5}$		1.30	1.30	1.31	1.30	= 1:1:1:1	[1]	
	Mg	O	H	Cl														
	$\frac{31.65}{24.3}$	$\frac{20.84}{16}$	$\frac{1.31}{1}$	$\frac{46.2}{35.5}$														
1.30	1.30	1.31	1.30	= 1:1:1:1														
MgOHCl	[1]	[2]																

75. M/J 15/P22/Q2/a(i),e

2	(a) (i)	$2\text{PbS} + 3\text{O}_2 \rightarrow 2\text{PbO} + 2\text{SO}_2$ reagents and formulae balancing	[1] [1]	[2]
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(e) (i)	$12.35 \times 0.01 / 1000 = 1.235 \times 10^{-4}$	[1]	[1]
(ii)	$1.235 \times 10^{-4} \times 1000 / 50 = 2.47 \times 10^{-3}$	[1]	[1]
(iii)	$2.47 \times 10^{-3} \times 64.1 = 0.158327 \text{ g} = 158 \text{ (3 sf only)}$	[1]	[1]

76. M/J 15/P23/Q1/c

(c) (i)	mean/average mass of the isotopes/an atom(s) relative to 1/12 of the mass of an atom of ^{12}C /on a scale where an atom of ^{12}C is (exactly) 12	[1] [1]	[2]
(ii)	$20.2 = \frac{(20 \times 90.48) + (21 \times 0.27) + (9.25y)}{100}$ $\frac{2020 - 1815.27}{9.25} = 22.133$ $y = 22$	[1] [1]	 [2]

77. M/J 15/P23/Q2/d(iii)

(iii)	$1 \times 10^{-4} \times 8000 = 0.8 \text{ mol H}^+$	[1]	
	$\frac{0.8}{2} \times 100.1 = \text{mass CaCO}_3 = 40 \text{ g}$	[1]	[2]